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ANTENNA**

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ក្រសួងអប់រំ យុវជន និងកីឡា វិទ្យាស្ថានបច្ចេកវិទ្យាកម្ពុជា



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សេចក្តីសង្ខេប

ការរចនាអង់តែនដែលមានទំហំតូច ទំន់ស្រាល និងចំណាយទាប កម្រិតបញ្ជូនមានប្រសិទ្ធភាពធំទូលាយ និងការសាយភាយក្រោមទំរង់ Omnidirectional ត្រូវបានលើកឡើងនៅក្នុងនិក្ខេបបទនេះ។ អង់តែនដែលបានលើកឡើងត្រូវបានផ្តុំឡើងដោយអង់តែនម៉ូណូប៉ូលពីរបន្ទះដែលមានរូបរាងជាអក្សរក្រិច Γ ផ្ដេក និងអក្សរអង់គ្លេស F។ ការរចនាអង់តែនដែលមានទំហំត្រឹមតែ $22 \times 42 \times 1.6 \text{ mm}^3$ ទទួលបានជោគជ័យដោយសារការដាក់បញ្ចូលនៃបន្ទះអង់តែនរាងអក្សរក្រិច Γ ផ្ដេកទៅក្នុងបន្ទះរាងអក្សរអង់គ្លេស F។ អង់តែនម៉ូណូប៉ូលពីរបន្ទះ និងប្លង់ខាងក្រោមរបស់វាត្រូវបានព្យាយាម និងបិទទៅនឹងផ្ទៃសងខាងនៃស្រទាប់ FR4។ ការភ្ជាប់បន្ទះខ្សែ Microstrip ធ្វើឡើងដើម្បីផ្គត់ផ្គង់ទៅដល់អង់តែន។ កម្រិតផ្សាយក្នុង multiband ត្រូវទទួលបានដោយការកំណត់រចនាសម្ព័ន្ធនៃបន្ទះម៉ូណូប៉ូលទាំងពីរ ខណៈដែលកម្រិតបញ្ជូនធំទូលាយនៅប្រេកង់ខ្ពស់ត្រូវទទួលបានដោយការកំណត់ចំងាយ Coupling រវាងបន្ទះទាំងពីរ។ ធៀបនឹងតម្លៃ S-Parameter ដែលតូចជាង -7.5 dB (ឬ Voltage Standing-Wave Ratio ដែលតូចជាង 2.5:1) 35.23% នៃប្រេកង់ 900 MHz និង 59.28% នៃប្រេកង់ 2200 MHz ទទួលបានដោយប្រើកម្មវិធី High Frequency Structure Simulator។ លើសពីនេះទៅទៀត Gain របស់អង់តែនក្នុងចន្លោះប្រេកង់ទាបមានតម្លៃពី 1.8812 dBi ទៅ 2.0257 dBi និងប្រេកង់ខ្ពស់មានតម្លៃពី 1.4941 dBi ទៅ 4.5661 dBi ជាមួយនឹងប្រសិទ្ធភាពនៃការផ្សាយខ្ពស់ជាង 70%។ លទ្ធផលនៃការពិសោធន៍ដោយប្រើឧបករណ៍វិភាគបណ្តាញ Agilent E5071C Series Network Analyzer បានផ្ទៀងផ្ទាត់ទៅនឹងលទ្ធផលដោយប្រើកម្មវិធី High Frequency Structure Simulator។ ជាលទ្ធផល អង់តែនដែលបានលើកឡើងក្នុងនិក្ខេបបទនេះអាចប្រើការបានក្នុងទូរស័ព្ទចល័តដែលប្រតិបត្តិដោយប្រព័ន្ធ GSM/UMTS/LTE/RFID/Zigbee/WLAN/Wi-Fi/Bluetooth។

RESUME

Une nouvelle antenne interne unipolaire imprimée pour les appareils mobiles modernes, qui a la taille compacte, légère, à faible coût, large bande passante de l'impédance, et diagramme de rayonnement omnidirectionnel, est proposée dans cette thèse. Mentionnée antenne unipolaire consiste à deux rubans, qui sont le ruban de l'horizontale- Γ et F-forme. En intégrant le ruban horizontal- Γ dans le ruban en forme de F, l'antenne proposée occupe un volume compact de seulement $22\text{mm} \times 42\text{mm} \times 1.6\text{mm}$. Antenne unipolaire à deux rubans et son plan de masse sont imprimés séparément sur la double surface du substrat FR4 en utilisant la technologie de circuit imprimé. Pour exciter l'antenne, méthode d'alimentation de la ligne de micro-ruban est utilisée. La multi bande est obtenue par la configuration à deux rubans tandis que la large bande passante à haute fréquence est attribuée au couplage mutuel entre ces deux rubans. Comme confirmé par High Frequency Structure Simulator, le Paramètre S qui est moins de -7.5dB ou équivalente à rapport d'onde stationnaire qui est inférieur à 2,5, les bandes passantes de l'antenne proposée sont environ 35,23% au 900 MHz et 59,28% au 2200 MHz. En outre, il est constaté que le gain de l'antenne est d'environ 1.8812dBi - 2.0257dBi à la bande de fréquences basse et 1,4941 - 4.5661dBi à la bande de fréquence haute avec une efficacité de rayonnement supérieure à 70%. En utilisant Agilent E5071C ENA Series Network Analyzer, les résultats expérimentaux de paramètre S et le rapport d'onde stationnaire vérifient les simulations. Par conséquent, l'antenne proposée peut être servie comme antenne interne pour l'application de téléphone portable qui fonctionne dans le domaine de GSM/UMTS/LTE/RFID/Zigbee/WLAN/Wi-Fi/Bluetooth.

ABSTRACT

A novel internal printed monopole antenna for modern mobile devices, which has compact size, lightweight, low cost, wide impedance bandwidth, and omnidirectional radiation pattern, is proposed in this thesis. The proposed antenna consists of two-strip monopole antenna, which are horizontal- Γ strip and F-shape strip. By embedding the horizontal- Γ strip into the F-shape strip, the proposed antenna occupies a compact volume of only 22mm $\times$ 42mm $\times$ 1.6mm. Two-strip monopole antenna and its ground plane are separately printed on difference side of FR4 substrate using printed circuit technology. Microstrip line feeding method is used to excite the antenna. The multiband is obtained by two-strip configuration while wide bandwidth at high frequency is attributed to the mutual coupling between these two strips. As confirmed by High frequency Structure Simulator for S-Parameters (i.e. Return Loss) less than -7.5dB or equivalently Voltage Standing-Wave Ratio less than 2.5, the fractional bandwidths of the proposed antenna are around 35.23% at 900MHz and 59.28% at 2200MHz. Moreover, it is found that antenna gain is about 1.8812dBi – 2.0257dBi at the low frequency band and 1.4941 – 4.5661dBi at the high frequency band with radiation efficiency higher than 70%. By using Agilent E5071C ENA Series Network Analyzer, the experimental results of S-Parameter and Voltage Standing-Wave ratio verify the simulations. Therefore, the proposed antenna can be served as internal antenna for GSM/UMTS/LTE/RFID/Zigbee/ WLAN /Wi-Fi/Bluetooth mobile applications.

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ABBREVIATIONS AND SYMBOLS

ABC	Radiation Boundary Condition
AUT	Antenna Under Test
BS	Bus station
CEM	Computational Electromagnetic
DCS	Digital Cellular System
FDM	Frequency-Division Multiplexing
FDMA	Frequency Division Multiple Access
FDTD	Finite-Difference Time Domain
FEM	Finite Element Method
GMSK	Gaussian Minimum Sift-Keying
GSM	Global System for Mobile communications
HFSS	High Frequency Structure Simulator
IF	Intermediate Frequency
ILA	Inverted-L Antenna
IMT	International Mobile Telecommunications
ISM	Industrial Scientific and Medical
ITU	International Telecommunication Union
LTE	Long Term Evolution
MoM	Moment of Method
MS	Mobile Station
OFDMA	Orthogonal Frequency Division Multiple Access
PCB	Printed Circuit Board
PEC	Perfect Electric Conductor
PIFA	Planar Inverted-F Antenna
PMA	Planar Monopole Antenna
PML	Perfect Matched Layer
QAM	Quadrature Amplitude Modulation
QPSK	Quadrature Phase Sift-Keying
RFID	Radio Frequency Identification
SAR	Specific Absorption Rate
SC-FDMA	Single Carrier Frequency Division Multiple Access
TDM	Time-Division Multiplexing

TDMA	Time Division Multiple Access
TEM	Transverse Electromagnetic
TM	Transverse Magnetic
UMTS	Universal Mobile Telecommunication System
VNA	Vector Network Analyzer
VSWR	Voltage Standing-Wave Ratio
WCDMA	Wideband Code Division Multiple Access
WLAW	Wireless Local Area Network
Z_c	Characteristic impedance of feed line
$\varepsilon_{r,eff}$	Effective dielectric constant of substrate
μ_0	Permeability in vacuum
ε_0	Permittivity in vacuum
ε_r	Relative permittivity or dielectric constant of material
Q	Quality factor
μ_r	Relative Permeability
c	Speed of light in vacuum ($2.99792458 \times 10^8 \text{ m/s}^2$)
λ	Wavelength
dB,dBi	Decibel, Decibel-isotropic

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1. INTRODUCTION

1.1. Evolution of Cellular Communication

As the grown of the number of cellular subscribers with the limitation of the electromagnetic spectrum, cellular communication systems, which use frequency reuse method, have been proposed (Mitra, 2009).

Typically, a cellular communication system is a 3-layer system, which contains the core, the edge, and the access subsystem (Toh, 2011). The core subsystem is the kernel of the network system, which perform switching, handoffs, signaling, traffic control and management, user profile, and interaction with the essential database. The edge subsystem provides the interface to connect the core subsystem to the access subsystem. The access subsystem (frequently called base transceiver station) provides the interface for connecting the network to the user equipment (Toh, 2011). Using multiplexing method such as Time-Division Multiplexing (TDM) or Frequency-Division Multiplexing (FDM) and full duplex technique like Time Division Duplexing or Frequency Division Duplexing allow Mobile Station (MS) to communicate with Base Station (BS) by two ways (downlink and uplink) simultaneous of communication (CIHANGIR, 2014).

First Generation (1G) is the first analog cellular mobile communications, which use Frequency Division Multiple Access (FDMA)/ FDD and analog Frequency Modulation (FM) were presented in Scandinavia and US in 1981 and 1983, respectively (Mitra, 2009).

Using digital modulation format, second Generation (2G), which use Time Division Multiple Access (TDMA)/FDD and Code Division Multiple Access (CDMA)/FDD, was introduced in the middle of the 1980s. Digital Cellular System (DCS), Personal Communication System (PCS), and Global System for Mobile Communication (GSM) network are presented in 2G. Particularly, GSM is one of the most commonly used system. It use TDMA standard that utilizes Gaussian Minimum Shift-Keying (GMSK) modulation technique for improves the spectral efficiency. Many GSM standards are used in difference region in the world. GSM850 use 824 – 849 MHz for uplink and 869 – 894 MHz for downlink while GSM900 use 935 – 960 MHz for uplink and 890 – 915 MHz frequency band for downlink. On the other hand, GSM1800 use 1710 – 1785 MHz for uplink and 1805 – 1880 MHz frequency band for and downlink while GSM1900 use 1850 – 1910MHz for uplink and 1930 – 1990 MHz for downlink.

At the beginning of the 21th century, International Mobile Telecommunications-2000 (IMT-2000) standard was released by International Telecommunication Union (ITU).

Universal Mobile Telecommunication System (UMTS) is one of IMT-2000 parts that have been agreed as third generation (3G) standard by Europe, Japan, and Asia. UMTS or Wide-Code Division Multiple Access (W-CDMA) uses 1710 – 2170 MHz frequency band for sending the information between MS and BS.

With the completed evaluation of 3G, Long Term Evaluation Advance (LTE-A) is accepted to be the fourth generation (4G) of the cellular communication systems. With this technology, Orthogonal Frequency Division Multiple Access (OFDMA) technique, and QPSK, 16QAM, and 64QAM modulation scheme are used for downlink communication while Single Carrier FDMA (SC-FDMA) technique and QPSK and 16QAM modulation scheme are used in uplink communication. Among many standard of LTE technology, LTE2300 use the frequency band from 2305 to 2400 MHz while LTE2500 use the frequency band from 2500 to 2690 MHz.

1.2. Common wireless Transmission Protocols

Zigbee is a specification for a suite of high-level communication protocol, which base on IEEE 802.15.4 standard. It provides the industrial wireless mesh networking for connecting the sensors, instrumentation, and control system (Somani & Patel, 2012). An application of Zigbee in cell phone is short-range radio wireless headphones (Mitra, 2009). Furthermore, Radio Frequency Identification (RFID) is wireless sensor modules that use electromagnetic field to transfer data for identifying and tracking tags attached to the objects. Even though Zigbee and RFID reader are used in difference purpose, they commonly operate in Industrial Scientific and Medical (ISM) band. In most of the world, 2.4 GHz frequency band have been used. In some regions such as USA and Australia, they operate at 860 – 956 MHz frequency band. Furthermore, Wireless-Fidelity (Wi-Fi)/Wireless Local Area Network (WLAN), which use IEEE 802.11b/g/n protocol, also operate at ISM2450 band (2400 – 2484 MHz) (Mitra, 2009).

1.3. Evolution of Mobile Phone Antenna

The evaluations of the cellular mobile communications have caused the demand for the advancement of the novel multiband antenna design. Through the end of the 1990s, external antenna like whip antenna or/and helical antenna had been mounted on the top of the mobile phone's chassis. The length of the whip antenna is quarter wavelength at the operating frequency. In order to reduce the size of the antenna, helical form is used and therefore name helical antenna. For operating in dual-band frequency, the combination of the whip antenna and helical antenna had been used. In spite of having high performance, external antenna has a

drawback of high Specific Absorption Rate (SAR). Therefore, the internal antenna is preferred. Internal antenna is release in 1998 and 1999 by Nokia Corporation. The major of the internal antenna that have been used in mobile terminal are Planar Inverted F Antenna (PIFA) and Planar Monopole Antenna (PMA) due to its compact size and versatile shape (Toh, 2011).

1.4. Problem Statement and Motivation of Thesis

As mentioned above, cellular communication systems have been growth very rapid. Since it is sophisticated and expensive to replace the whole system by the novel system, the original systems are therefore upgraded to the next cellular communication generation. As a result, the novel antennas have been designed for supporting not only the frequency band that operates at original cellular communication systems but also the ever-increasing implement of the novel frequency band (i.e. 4G frequency band or further). Moreover, some other frequency bands that operate in small coverage are also indispensable for modern smart phone. Therefore, multiband and wideband antennas are preferred in the modern mobile terminal.

1.5. Scope of Thesis

The aim of this thesis is to design a microstrip-fed multiband and wideband printed monopole antenna for modern mobile devices whose operating frequency band are listed in Table 1.1. As the requirement of mobile applications, the proposed antenna should has a compact-size, low-cost, lightweight, omnidirectional radiation pattern, and high impedance bandwidth.

1.6. Objective of Thesis

The objectives of thesis are:

- To design of a microstrip-fed multiband and wideband printed monopole antenna, which compose of a horizontal- Γ strip and an F-shape strip. Parametric optimizations are need to be involved for obtaining a compact-size and broadband antenna using Ansoft High Frequency Structure Simulator (HFSS) software package.
- To fabricate the designed antenna on a low-cost substrate using Printed Circuit Board (PCB) technique.
- To measure the S-Parameter (Return Loss) and Standing-Wave Ratio (SWR) of fabricated antenna using Agilent E5071 ENA Series Network Analyzer.
- To compare between simulated results and experimented results of S-Parameter and SWR of the proposed antenna.

Table 1.1. Wireless application and correspond frequency band

Wireless Application	Frequency Band
GSM850	824 – 894 MHz
GSM900	890 – 960 MHz
GSM1800	1710 – 1880 MHz
GSM1900	1850 – 1990 MHz
UMTS	1920 – 2170 MHz
LTE2300	2305 – 2400 MHz
LTE2500	2500 – 2690 MHz
Zigbee (IEEE 802.15.4)/RFID	868 – 870 MHz 902 – 928 MHz 2200 – 2300 MHz 2400 – 2484 MHz
Wi-Fi/WLAN (IEEE 802.11b/g/n)	2400 – 2484 MHz

1.7. Organization of Thesis

To provide the reader to understand about the proposed antenna, it is necessary to start from some antenna basics and fundamental parameters, which will be presented in section 2. To obtain the accuracy result of the antenna parameters using numerous simulation, some specifications and design procedures of HFSS software package are also introduced in section 3.1. Importantly, the design of the proposed antenna and the realized antenna, and Vector Network Analyzer measurement will be presented in section 3.2 and 3.3, respectively. In addition, the comparison between simulated and experimented results will be discussed in section 4. At the end of this thesis, the conclusions and further recommendations will be introduced in section 5.

2. LITERATURE REVIEW

2.1. History and Definition of Antenna

In 1873, James Clerk Maxwell published his unified theory of electricity and magnetism, and therefore proved the existence of electromagnetic wave that propagate in free space at the speed of light ($c=2.99792458 \times 10^8 \text{ m/s}$). After that, Professor Heinrich Rudolph Hertz had demonstrated the confirmation of Maxwell's theory and reality in 1886. Professor H. R. Hertz was capable to produce a spark of 4m wavelength in the gap of transmitter $\lambda / 2$ dipole antenna, which was then detected as a spark in the gap of a nearby loop (Balanis, 2005). After that time, antenna make possible of wireless communication and therefore have been developed until present.

According to *IEEE Standard Definitions of Terms for Antennas* (IEEE Std 145-1993, 1993), define *antenna* or *aerial* as "That part of a transmitting or receiving system that is designed to radiate or to receive electromagnetic waves." In other words, antenna is defined as a transducer between a guided wave propagating in a transmission line and an electromagnetic wave propagating in an unbounded medium (usually free space), or vice versa. In addition, Figure 1.1 shows a transmitter system that contain antenna as a transition device.

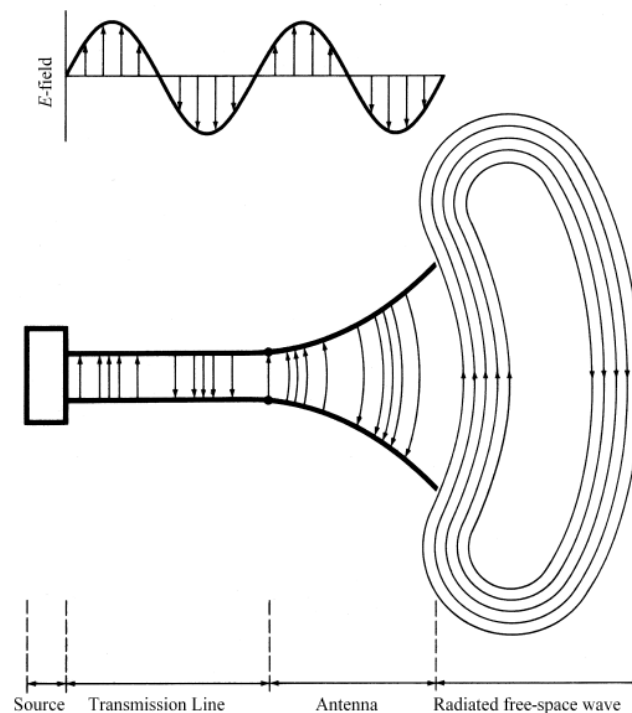


Figure 1.1. Antenna as a transition device (Balanis, 2012)

2.2. Fundamental Parameters of Antennas

In term of trade-off between radiation performances and impedance bandwidth of antenna, antenna size reduction is restricted by fundamental physical limits (Huitema & Monediere, 2014). Miniaturization of antenna size with high performance is hard to achieve. As a result, it should be understand about the characteristics of the antenna, i.e. antenna parameters, which will be described as follow (Gustanfsson & Jonsson, 2015). One of the most kernel parameters of the antenna is resonance phenomenon. Base on *IEEE Standard Definition of Term for Radio Wave Propagation* (IEEE Std 211-1997, 1997), *resonance* of a traveling wave is defined as “The change in magnitude as the frequency of the wave approaches or coincides with a natural frequency of the medium.”

2.2.1. Impedance, Bandwidth, Quality Factor and Scattering Parameters of antenna

The literature that will be described about Impedance, Bandwidth, and Quality factor of antenna below is referred to (Yaghjian & Best, 2005).

①. Input Impedance of Antenna

From Figure 1.1, considering the antenna that consist of linear materials fed by transmission line or waveguide (i.e. feed line) that carries just only one mode of propagation at time-harmonic ($e^{j\omega t}$) frequency $\omega > 0$. The input impedance $Z(\omega)$ of tune antenna terminal from arbitrary terminal in the feed line to the end of antenna that compose of resistance $R(\omega)$, which is the combination of radiation resistance (R_R) and antenna loss (R_L), and reactance $X(\omega)$ can be expressed as

$$Z(\omega) = R(\omega) + jX(\omega) = \frac{V(\omega)}{I(\omega)}, \quad V(\omega) = a(\omega) + b(\omega), \quad I(\omega) = \frac{a(\omega) - b(\omega)}{Z_c} \quad (\text{Eq. 1})$$

where Z_c Characteristic impedance of feed line. In practice, the value of characteristic impedance of transmission line are typically 50Ω or 75Ω .

$a(\omega), b(\omega)$ Amplitude of incident, reflection wave within feed line.

$V(\omega), I(\omega)$ Voltage, current at antenna input terminal.

Particularly, the average stored electric ($\overline{W_e}$) and magnetic energies ($\overline{W_m}$) at resonance frequency are equal and therefore $X_0(\omega_0) = 0$ and $X_0'(\omega_0) = \frac{dX(\omega_0)}{d\omega} > 0$.

From above equation, it can be observed that the Impedance of antenna is not depends only on its geometry, method of excitation and proximity to surround objects but also depend on the frequency. It is mean that the antenna can be matched to the connected transmission line and other associated equipment only a fractional of frequency band. As a result, antenna impedance is the kernel parameter that characterizes the useful bandwidth of antenna. In addition, it also exhibit the relationship between transferred power from the source and accepted power by the antenna (IEEE Std 149-1979, 1979).

②. Bandwidth Formulas of Antenna

Since antenna impedance bandwidth describes the frequency range, over which the antenna can be properly radiated or received energy, it is necessary to characterize the performance of antenna in term of its bandwidth.

Conductance bandwidth and matched Voltage Standing-Wave Ratio (VSWR) bandwidth are the two ways to define the bandwidth of an antenna, which is tuned to resonance frequency or equivalent to zero reactance. Since the expression of conductance bandwidth cannot apply for all frequency range, it is better to define the fundamental bandwidth of the antenna in term of matched VSWR bandwidth. Therefore, only matched VSWR bandwidth will be presented in the following section.

For tuned antenna, the fractional matched VSWR bandwidth $FBW_V(\omega_0)$ is defined as the difference between the two frequencies on either side of resonance frequency ω_0 at which $s = \text{VSWR}$, or, equivalently, at which the magnitude squared of the reflection coefficient $\alpha = |\Gamma_0(\omega)|^2 = (s - 1)^2 / (s + 1)^2 \leq 0.5$, provide the characteristic impedance Z_c of the feed line equals to $Z_0(\omega_0) = R_0(\omega_0)$, which is expressed as

$$FBW_V(\omega_0) = \frac{\omega_+ - \omega_-}{\omega_0} \simeq \frac{4\sqrt{\beta}R_0(\omega_0)}{\omega_0|Z_0'(\omega_0)|}, \quad \sqrt{\beta} = \sqrt{\frac{\alpha}{1-\alpha}} = \frac{s-1}{2\sqrt{s}} \leq 1 \quad (\text{Eq. 2})$$

Where ω_0	Resonance frequency
ω_-, ω_+	Lower, upper frequency of resonance frequency ω_0
s	Arbitrary VSWR value (Typically 2 : 1, 2.5 : 1 and 3 : 1)
$Z_0'(\omega_0)$	The derivative of antenna input impedance, which is tuned to the resonance frequency

The approximation in (Eq. 2) is satisfied when $\Delta\omega_{\pm} / \omega_0 \ll 1$.

In the case of half-power VSWR bandwidth, $\alpha = 0.5$ ($s = 5.828$) and $\beta = 1$. Typically, for antenna that has fractional bandwidth higher than 50% can be considered as wideband antenna.

③. Quality factor or Q and its relation to bandwidth of antenna

For antenna, which is tuned to obtain zero reactance at frequency ω_0 ($X(\omega_0) = 0$), the quality factor $Q(\omega_0)$ can be expressed in term of total energy $W(\omega_0)$ and accepted energy by antenna as

$$Q(\omega_0) = \frac{\omega_0 |W(\omega_0)|}{P_A(\omega_0)} \quad (\text{Eq. 3})$$

Particularly, $P_A(\omega_0) = |I_0|^2 R_0 / 2$ and $W(\omega_0) = |I_0|^2 X_0'(\omega_0) - [W_L(\omega_0) - W_R(\omega_0)]$, $Q(\omega_0)$ can be written as

$$Q(\omega_0) = \left| \frac{\omega_0}{2R_0(\omega_0)} X_0'(\omega_0) - \frac{2\omega_0}{|I_0|^2 R_0(\omega_0)} [W_L(\omega_0) - W_R(\omega_0)] \right| \quad (\text{Eq. 4})$$

Where $R_0(\omega_0)$ Resistance of tuned antenna at resonance frequency ω_0

$X_0'(\omega_0)$ Derivative of tuned antenna reactance at resonance frequency ω_0

I_0 Input current enter the tuned antenna at resonance frequency ω_0

$W_L(\omega_0), W_R(\omega_0)$ Dispersive quantities associated with the power dissipated as power loss and radiated power by antenna.

To characterize antenna's bandwidth, the inverse relationship between fractional matched VSWR bandwidth and $Q(\omega_0)$ are necessary, by approximation yield

$$Q(\omega_0) \simeq \frac{\omega_0}{2R_0(\omega_0)} |Z_0'(\omega_0)| \simeq \frac{2\sqrt{\beta}}{FBW_V(\omega_0)}, \quad \sqrt{\beta} = \frac{s-1}{2\sqrt{s}} \leq 1 \quad (\text{Eq. 5})$$

This expression introduces the trade-off relationship between antenna Q and fractional matched VSWR bandwidth. In other words, high value of Q will increase the radiation efficiency but narrow the bandwidth. In design, the optimal value of Q with appropriate bandwidth may be desired.

④. Scattering Parameters of Antenna

In application, a microwave network may contains multi-port. It is usually much easier to apply circuit analysis to a microwave problem than it is to solve Maxwell's equation

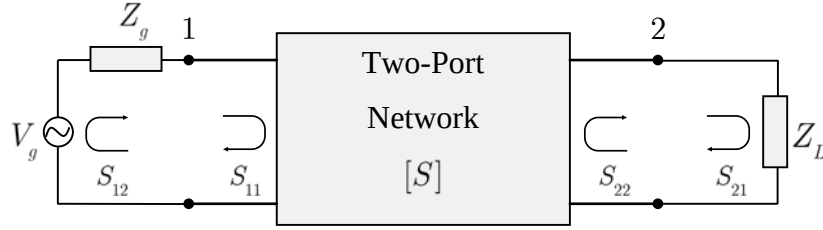


Figure 2.2. Two-Port Microwave Network

(Appendices) for the same problem. This is because Maxwell's equation solves all solution in all point in space while only the some parameters such as voltage or current at a set of terminals or port or power flow through a device etc. is interested in real application (Pozar, 2011). As a result, the microwave network analysis is preferred. In addition, it can be used to analyze the performance of the antenna in term of Scattering Parameter.

By driving the power to network and measuring the reflected power from the network, the characteristic of the network can be investigated. In addition, Q -factor and therefore impedance bandwidth of the antenna are related to the accepted power by the antenna. Therefore, network analysis capable to characterize the power ration between incident and reflection wave at an operating frequency rang. In commercial software package, such as High Frequency Structure Simulator (HFSS) and measurement devices such Vector Network Analyzer (VNA), allow the calculation the value of the S-Parameter to investigate the performance of the antenna. As a result, network analysis can be taken into account the calculation of impedance bandwidth of the antenna in theory, simulation, and measurement.

There are many network analysis method that involve in analysis of microwave network such as impedance and admittance matrices, Scattering matrix etc. For non-TEM transmission line, a practical problem exists when trying to measure voltages and currents at microwave frequency since direct measurement usually involve the magnitude and phase of a wave traveling in a given direction or of a standing wave. Thus equivalent voltages and currents, and related impedance and admittance matrices, become somewhat of an abstraction when dealing with high-frequency networks. A representation more in accord with direct measurement, and with the ideas of incident, reflected, is given by the scattering matrix.

Considering a two-port microwave network shown in Figure 2.2, where a is the amplitude of the voltage wave incident on port n ($n=1,2$) and b is the amplitude of the voltage wave reflected from port n , the scattering matrix, or $[S]$ matrix, is expressed in term of incident and reflected wave as

$$\begin{bmatrix} b_1 \\ b_2 \end{bmatrix} = \begin{bmatrix} S_{11} & S_{12} \\ S_{21} & S_{22} \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \end{bmatrix} \quad \text{Or equivalently } [b] = [S][a] \quad (\text{Eq. 6})$$

were $S_{11}, S_{12}, S_{21}, S_{22}$ are the Scattering Parameters or S-Parameters of the Scattering matrix or S-matrix. In term of incident and reflection wave magnitude, a_1, a_2, b_1, b_2 , the S-Parameters can be expressed as

$$S_{11} = \frac{b_1}{a_1}, \quad S_{12} = \frac{b_1}{a_2}, \quad S_{21} = \frac{b_2}{a_1}, \quad S_{22} = \frac{b_2}{a_2} \quad (\text{Eq. 7})$$

In other words, S_{11} (S_{22}) are the reflection coefficients, seen looking into ports 1 (2) when all others ports are terminated in matched load to avoid reflection. S_{12} (S_{21}) is the transmission coefficients characterized power that transfer from port 1 (2) to port 2 (1) when all other ports are terminated in matched loads. When S_{12} equal to S_{21} , the network is thus called reciprocal.

In design that will be processed in antenna design in the Section 3, only $S_{11} = \Gamma_{in} = \frac{Z_{in} - Z_c}{Z_{in} + Z_c}$

that will be utilized to execute the impedance bandwidth of antenna. It is convenient to expressed the S_{11} in scalar logarithmic (decibel or dB) expression, which is called input non-negative Return Loss (RL) can be expressed as

$$RL = 10 \log_{10} \left(\frac{1}{|S_{11}|^2} \right) = -20 \log |S_{11}| \quad |S_{11}| \leq 1 \quad (\text{Eq. 8})$$

Commonly, return loss that greater than 6dB, 7.5dB and 10dB, which are correspond to the VSWR ratio of 3:1, 2.5:1 and 2:1 respectively, are typically chosen in antenna design. In addition, 7.5dB of return loss is chosen in the design that will be followed in the section 3.

2.2.2. Antenna Radiation Patterns

The *IEEE Standard Definitions of Terms for Antennas* (IEEE Std 145-1993, 1993) define the *radiation pattern* or *antenna radiation* as “The spatial distribution of a quantity that characterizes the electromagnetic field generated by an antenna.” Before describing the radiation properties of antenna, radiation geometries will categorize the radiation pattern types, which will be introduced as follow.

- Radiation Lobes is a “portion of the radiation pattern bounded by regions of relatively weak radiation intensity,” (IEEE Std 145-1993, 1993). Figure 2.3 (a) shows the

radiation lobes that contain major, minor, side, and back lobes of an antenna pattern. Field distribution of radiation pattern of antenna can be categorized as isotropic, directional, and omnidirectional pattern. An isotropic radiator is defined as “A hypothetical, lossless antenna having equal radiation intensity in all directions.” To characterize the performance of the antenna radiation patterns, an isotropic radiator is usually utilized as the reference for expressing the directive properties of actual antennas. In contrast to isotropic antenna, omnidirectional antenna is defined as “an antenna having an essentially non-directional pattern in a given plane of the antenna and a directional pattern in any orthogonal plane.” On the other hand, a directional antenna is defined as “an antenna having the property of radiating or receiving electromagnetic waves more effectively in some directions than others” (IEEE Std 145-1993, 1993).

- Field Region of antenna is usually subdivided into three region: reactive near field, radiating near field (Fresnel) and far field (Fraunhofer) region (Balanis, Antenna Theory Analysis and Design, 2005). The dimension of surrounding field region of antenna is shown in Figure 2.3 (b).

The following session will present the quantities that are mostly used to characterize the radiation of an antenna.

①. Radiation Power Density and Radiation Intensity

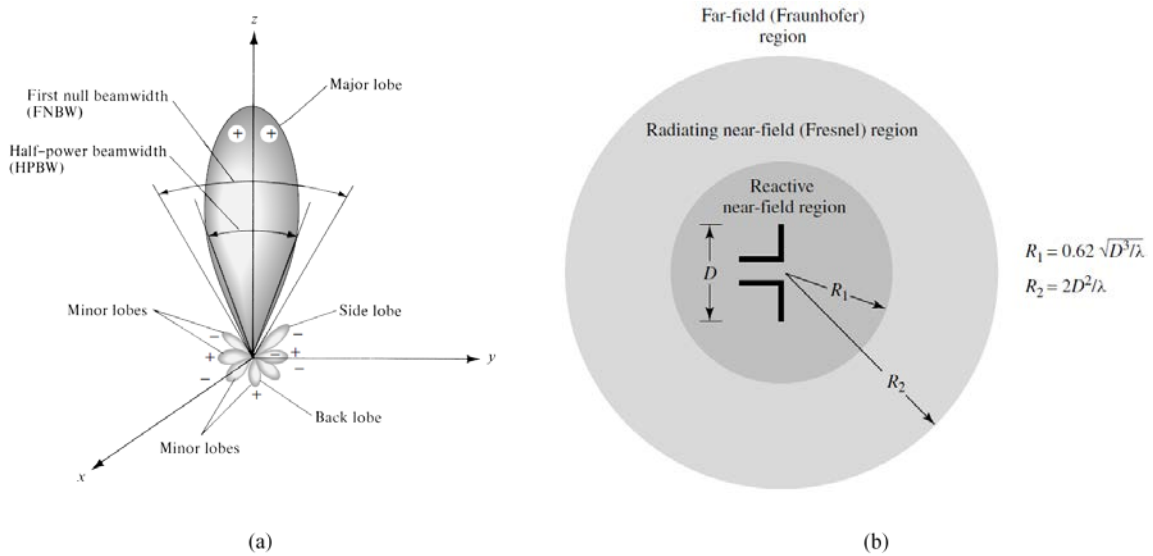


Figure 2.3. (a) Radiation lobes and beamwidth, and (b) field region of an antenna

(Balanis, 2012)

To communicate through wireless channel, the powers that associate with electromagnetic waves contain the desired information. Fortunately, Poynting vector can describe the power that associate with electromagnetic wave, which make possible of information transmission. In time-harmonic form, the time average Poynting vector can be expressed in term of time-harmonic electric and magnetic field as

$$W_{av}(x, y, z) = \frac{1}{2} \text{Re}[\vec{E} \times \vec{H}^*] \quad [W / m^2] \quad (\text{Eq. 9})$$

The above equation contain pure real part which mean that the average Poynting vector represent the power that is radiated by antenna. Therefore, the average power radiated by antenna enclosed by a surface (S) with a normal vector $\vec{n}$ can be written as

$$P_{rad} = P_{av} = \oint_S \vec{W}_{rad} \cdot d\vec{s} = \oint_S \vec{W}_{av} \cdot \vec{n} \, ds = \frac{1}{2} \oint_S \text{Re}[\vec{E} \times \vec{H}^*] \cdot d\vec{s} \quad (\text{Eq. 10})$$

In case of isotropic radiator, the relationship between Poynting vector and radiated power are given by

$$\vec{W}_0 = \left(\frac{P_{rad}}{4\pi r^2} \right) \vec{a}_r \quad [W/m^2] \quad (\text{Eq. 11})$$

Which is uniformly distributed over the surface of a sphere of radius r as the definition of isotropic radiator, which had mentioned above.

(IEEE Std 145-1993, 1993) define the *radiation intensity* of antenna as “in a given direction, the power radiated from an antenna per unit solid angle.” It is expressed a

$$U = r^2 W_{rad} \quad (\text{Eq. 12})$$

②. Antenna Gain and Radiation Efficiency

Gain is one of the most useful parameters in representing the performance of the antenna. In antenna specification sheet, antenna gain is commonly specified since it takes into account the actual loss of antenna. In addition, “gain does not include losses arising from impedance mismatches and polarization mismatches.” *Gain* (in a given direction) is define as “The ratio of the radiation intensity, in a given direction, to the radiation intensity that would be obtained if the power accepted by the antenna were radiated isotropically,” (IEEE Std 145-1993, 1993). In mathematically can be written as (Balanis, 2005)

$$\text{Gain} = 4\pi \frac{\text{radiation intensity}}{\text{total input (accepted) power}} = 4\pi \frac{U(\theta, \phi)}{P_{in}} \quad (\text{Eq. 13})$$

From above equation, antenna gain is a dimensionless quantity, which is a function of direction (i.e. angle). In the case that the direction is not specified, the direction of maximum

radiation intensity is implied (IEEE Std 145-1993, 1993). In transmitting mode, antenna gain shown how well of an antenna that will be able to convert the accepted power to electromagnetic wave in specified direction. Similarly, using reciprocity theorem, in receiving mode, antenna gain show how well of an antenna that is capable to convert received power in a given direction into electrical power.

In most applications, relative gain is more preferable than gain. (IEEE Std 145-1993, 1993) define *relative gain* as “The ratio of the gain of an antenna in a given direction to the gain of a reference antenna.” A dipole, horn, or any other antenna whose gain can be determined or known is usually used as the reference antenna. Particularly, the reference antenna is a lossless isotropic source, which can be written as (Balanis, 2005)

$$G = \frac{4\pi U(\theta, \phi)}{P_{in}(\text{lossless isotropic source})} \quad (\text{Eq. 14})$$

For convenient, gain would be expressed in logarithmic scale (decibel or dB and decibel-isotropic or dBi) as

$$G_{\text{dBi}} = G_{\text{dB}} = 10 \log_{10}(G) \quad (\text{Eq. 15})$$

Where dBi is just used to emphasize that this is the gain according to the definition, in which the antenna gain is compared to the reference antenna, i.e. isotropic radiator. Particularly, antenna gain of 3dBi means that the power that transmitting (or receiving) is 3dBi higher than that of the isotropic radiator with the same power in a given direction.

In addition to antenna gain, radiation efficiency is the indispensable parameter of antenna. *IEEE Standard Definitions of Terms for Antennas* (IEEE Std 145-1993, 1993) define *radiation efficiency* of antenna as “The ratio of the total power radiated by an antenna to the net power accepted by the antenna from the connected transmitter.” In other words, radiation efficiency is a measure of the efficiency at which the antenna converts the accepted power into the electromagnetic wave propagating to unbounded medium. In equation form, it can be expressed as (Balanis, 2005)

$$e_{cd} = \frac{P_R}{P_A} = \frac{R_R}{R_R + R_L} \quad (\text{Eq. 16})$$

Where R_R and R_L are the radiation resistance and conduction-dielectric loss of antenna, respectively.

From above equation, it can be noted that the radiation efficiency of antenna does not take into account the impedance mismatch between the feed line and antenna, and the miss polarization between the transmitting antenna and receiving antenna.

2.3. Microstrip Line

To excite the antenna, many different configurations can be used to feed the antenna. Particularly, microstrip line feeding method is one of the most useful planar transmission line feeding methods that usually involve in antenna feeding methods. Since it is easy to fabricate, simple to match to the antenna by varying the strip width, microstrip line feeding method is frequently chosen as the feeding method for exciting the planar antenna. Furthermore, it is also served as the feeding method for exciting the proposed antenna, which will be presented in the next chapter.

Figure 2.4 (a) shows a microstrip line, which consists of a conducting strip of width w and thickness t separated from the ground plane by an insulator or dielectric layer, called substrate, of dielectric constant ϵ_r and thickness h . In addition, since there is a separation between the substrate and air, there are three kinds of fields contained in the proximity of the microstrip line, which are the field in the substrate, the field in the air, and the field between the air and substrate called fringing field as shown in Figure 2.4. Therefore, a microstrip line cannot support a pure Transverse Electromagnetic (TEM) field configuration. To find the field configuration of the microstrip line within the substrate, the cavity model is one of the most useful models in modeling microstrip lines. To model a microstrip line by a cavity model, the microstrip line is approximated by the structure in Figure 2.4 (b). In addition, according to (Balanis, 2005), the four side walls of the microstrip line can be modeled as perfect magnetic conducting surfaces and the conducting strip and ground can be modeled as perfect electric conducting surfaces. Therefore, only TM^z field configurations exist within the substrate. By solving the waves that exist in the substrate from equations that are expressed in Appendices, the resonance frequency within the substrate can be written as

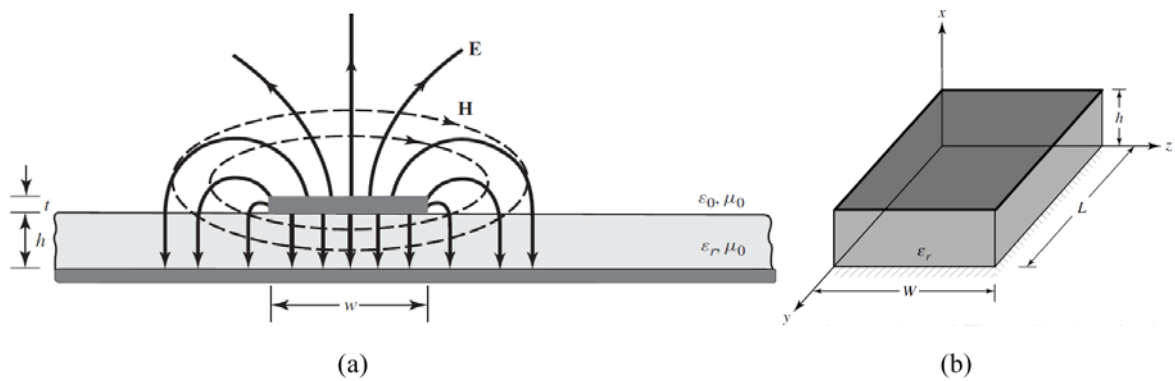


Figure 2.4. (a) Microstrip lines and (b) rectangular substrate integrated waveguide
(Balanis, 2005)

$$(f_r)_{mnp} = \frac{c}{2\pi\sqrt{\mu_r\epsilon_r}} \sqrt{\left(\frac{m\pi}{h}\right)^2 + \left(\frac{n\pi}{L}\right)^2 + \left(\frac{p\pi}{W}\right)^2} \quad \begin{matrix} m, n, p = 0, 1, 2, \dots \\ m = n = p \neq 0 \end{matrix} \quad (\text{Eq. 17})$$

Where c , μ_r and ϵ_r are the speed of light in vacuum ($c = 2.99792458 \times 10^8 \text{ m/s}$), relative magnetic permeability, and relative electric permittivity of the substrate, respectively. In the case of FR4 substrate, μ_r is approximately 1 while ϵ_r lie between 4.35 and 4.7.

Since there is the fringing field from the surface of the conducting strip to the ground plane as shown in Figure 2.4, it is necessarily to take it account into the fringing field into above resonance frequency expression. The air and substrate can be combined into a single quantity called effective dielectric constant. At low frequency, effective dielectric constant and effective width can be written as (Balanis, 2012)

$$\begin{aligned} - \frac{w_{eff}(0)}{h} &\leq 1 \\ \epsilon_{r,eff}(0) = \epsilon_{r,eff}(f=0) &= \frac{\epsilon_r + 1}{2} + \frac{\epsilon_r - 1}{2} \left\{ \left[1 + 12 \frac{h}{w_{eff}(0)} \right]^{-1/2} + 0.04 \left[1 - \frac{w_{eff}(0)}{h} \right]^2 \right\} \end{aligned} \quad (\text{Eq. 18})$$

$$\begin{aligned} - \frac{w_{eff}(0)}{h} &> 1 \\ \epsilon_{r,eff}(0) = \epsilon_{r,eff}(f=0) &= \frac{\epsilon_r + 1}{2} + \frac{\epsilon_r - 1}{2} \left[1 + 12 \frac{h}{w_{eff}(0)} \right]^{-1/2} \end{aligned} \quad (\text{Eq. 19})$$

Where

$$\begin{aligned} \left\{ \begin{aligned} \frac{w_{eff}(0)}{h} &= \frac{w_{eff}(f=0)}{h} = \frac{w}{h} + \frac{1.25t}{\pi h} \left[1 + \ln \left(\frac{2h}{t} \right) \right] \\ \frac{w_{eff}(0)}{h} &= \frac{w_{eff}(f=0)}{h} = \frac{w}{h} + \frac{1.25t}{\pi h} \left[1 + \ln \left(\frac{4\pi w}{t} \right) \right] \end{aligned} \right\} \quad \begin{matrix} \frac{w}{h} \geq \frac{1}{2\pi} \\ \frac{w}{h} < \frac{1}{2\pi} \end{matrix} \end{aligned} \quad (\text{Eq. 20})$$

In the cases that the thickness of conducting strip is very small compared to the high of the substrate ($t \ll h$), the effective width of conducting strip is equivalent to the physical width of conducting strip ($w_{eff} = w$).

In addition to the effective dielectric constant and effective width of microstrip line, the characteristic impedance of the microstrip line can be expressed as (Balanis, 2012)

$$\left\{ \begin{array}{l} Z_c(f=0) = \frac{60}{\sqrt{\varepsilon_{r,eff}(0)}} \ln \left[\frac{8h}{w_{eff}(0)} + \frac{w_{eff}(0)}{4h} \right] \\ Z_c(f=0) = \frac{120p}{\sqrt{\varepsilon_{r,eff}(0)} \left[\frac{w_{eff}(0)}{h} + 1.393 + 0.667 \ln \left[\frac{w_{eff}(0)}{h} + 1.444 \right] \right]} \end{array} \right. \quad \begin{array}{l} \frac{w_{eff}(0)}{h} \geq 1 \\ \frac{w_{eff}(0)}{h} < 1 \end{array} \quad (\text{Eq. 21})$$

2.4. Quarter-Wavelength Monopole Antenna

Half-wavelength dipole is one of the most commonly use antenna since its radiation resistance is 73Ω , which is closely to the characteristic impedance of the common transmission line. On the other hand, by cutting the under half of the dipole antenna and replacing by the ground plane, the monopole antenna is obtained as shown in Figure 2.5 (a). In addition, quarter-wavelength monopole and its image, by image theorem, form a half-wave dipole that radiates only the upper half of space as shown in Figure 2.5 (b). The Bandwidth of monopole antenna can be increased by increasing diameter. Furthermore, monopole antenna has 3dBi gain higher than that of the dipole antenna. Therefore, monopole antenna is more preferred than the dipole antenna in mobile terminal due to high directivity and compact size. There are many methods to calculation the length of the monopole, but most of them are very complicate to solve. Therefore, approximate formulas for rapid calculation that have high accuracy result are

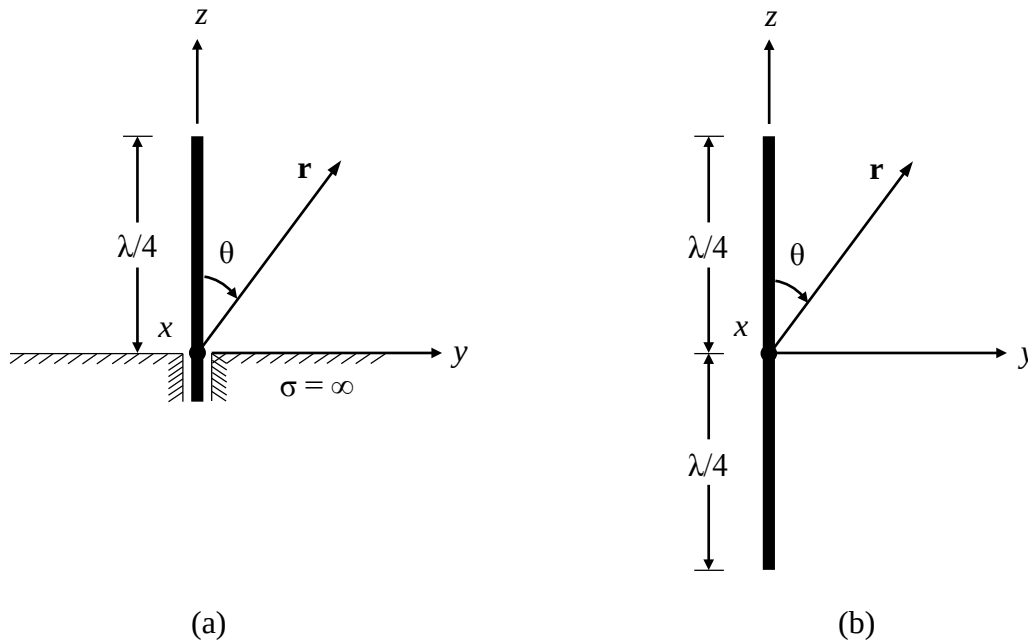


Figure 2.5. (a) Quarter-wavelength monopole on infinite electric conductor; (b) Equivalent of quarter-wavelength monopole on infinite electric conductor (Balanis, 2005).

sufficient for determining the length of the antenna in design procedure. This approximate formula define G of monopole antenna as (Balanis, 2005)

$$G = kl / 2 \quad \textbf{(Eq. 22)}$$

where l is the total length of the monopole antenna. It has been shown that the input resistance of the monopole antenna can be determined approximately as

$$\begin{array}{l} \hline 0 < G < \pi / 4 \\ \hline \text{(maximum input resistance of monopole is less than } 12.337\Omega) \\ R_{in} = 10G^2 \quad 0 < l < l / 8 \\ \hline \pi / 4 \leq G < \pi / 2 \\ \hline \text{(maximum input resistance of monopole is less than } 38.1915\Omega) \quad \textbf{(Eq. 23)} \\ R_{in} = 12.35G^{2.5} \quad l / 8 \leq l < l / 4 \\ \hline \pi / 2 \leq G < 2 \\ \hline \text{(maximum input resistance of monopole is less than } 100.265\Omega) \\ R_{in} = 5.57G^{4.17} \quad l / 4 \leq l < 0.3183l \end{array}$$

In addition to linear wire monopole antenna, planar monopole antenna can be equated to cylindrical monopole antenna with a large effective diameter (Kumar & Ray, 2003).

3. METHODOLOGY

3.1. Materials for the Study

3.1.1. Introduction to High Frequency Structure Simulator

As the increasing of the research and development of the wide variety of the real world application, a single component may contain many materials, which are dielectric, conductor, semiconductor, superconductor, ferrite material, metamaterial, etc. In this case, applying Maxwell's equation to the field problem is very complicated to determine or even though cannot lead to the final solution (Zhang & Sarkar, 2009). Therefore, a numerical method, well known as *Computational Electromagnetic* (CEM), has been proposed for solving the electromagnetic field problem. In order to solve the electromagnetic field problem, there are two basic steps of CEM, which are:

- ①. Discretizing Maxwell's integral or differential equation into a matrix equation by using Moment of Method (MoM), Finite Element Method (FEM), Finite-Difference Time Domain (FDTD), and so on.
- ②. Storing value and solving the unknown of the element of the matrix equation, by following the algorithm such as Higher-order basis function, LU factorization and so on.

The right technique for solving the problem should be chosen in order to obtain the accuracy result in a short executing period during the design procedure. Therefore, High Frequency Structure Simulator is chosen to simulate the proposed antenna in this design.

High Frequency Structure Simulator (HFSS) is an interactive software package, which is used for calculating electromagnetic behavior of a structure. The simulation technique that is used in HFSS is based on FEM, a method where the user defined 3D model is subdivided into small piece of finite elements, in order to discretize integral form of Maxwell's equation into matrix form. In addition, the finite elements that are used in HFSS is tetrahedra, and the combination of those tetrahedra are referred to finite element mesh (Ansoft, 2009). By using higher-order basis function algorithm, a solution is found in HFSS for the fields within the finite elements, and these fields are related to each other so that Maxwell's equations are

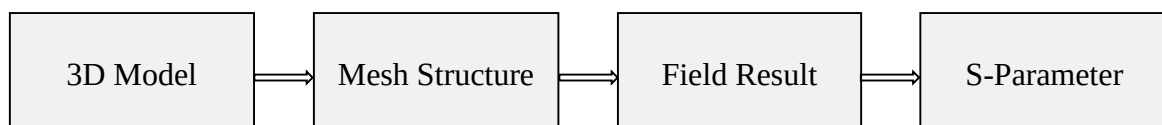


Figure 3.1. Block diagram of HFSS solving procedure (Ansoft, 2009)

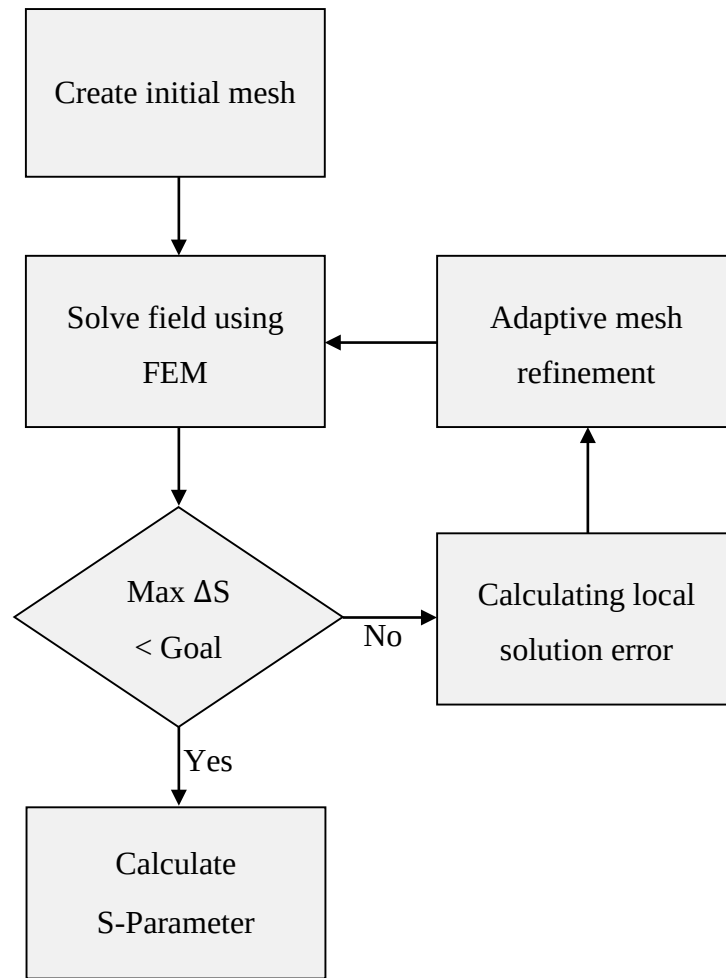


Figure 3.2. HFSS adaptive solution process (Ansoft, 2009)

satisfied across inter-element boundaries. Once the field problem is solved, the S-Parameter is obtained and the full wave analysis can be set-up and plotted. Figure 3.1 show the procedure that is used for solving the field problem in HFSS.

Figure 3.2 show The interactive of HFSS that make it reliable in solving electromagnetic field structure is that it uses adaptive solution process as shown in. Adaptive solution process is a process in which the region that has high error of electric field solution is redefined iteratively until the convergence criteria are satisfied or number of iterates is reached the maximum number that is defined by the user.

3.1.2. HFSS Design Procedure

To achieve the solution of electromagnetic field problem of a structure, six general steps must be involved in a proper HFSS simulation. Before working with these six steps, the solution setup must be specified first as recommended by software. Wrong solution type may lead to inaccuracy and invalid result. Three solution types are available in HFSS. The solution

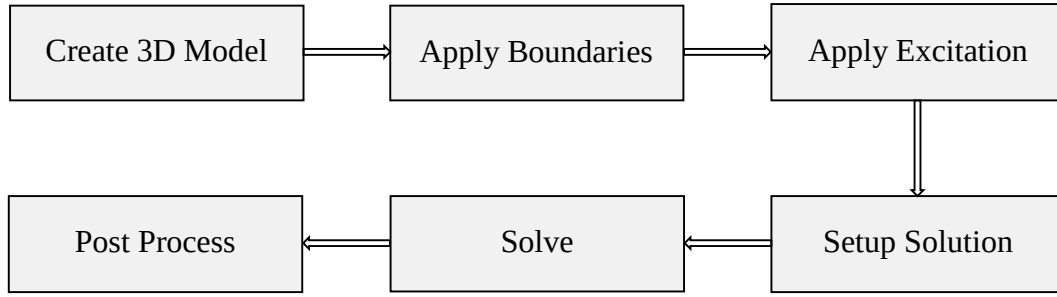


Figure 3.3. HFSS's six general steps in a simulation (Ansoft, 2009)

should be chosen base on the geometry that user prefer HFSS to solve. In the proposed antenna design, since antenna is excited via microstrip feed line, driven terminal solution type is the most convenient solution type, which is specified. After specifying the solution type, the following steps can be processed and flowed as Figure 3.3.

①. Create 3D model

Create the model or geometry that user want HFSS to analyze and solve. It should be note that not all of the geometries and element connection are valid. In addition, some little difference between the connections of the same geometries may cause a little difference in result. Using rectangular sheet (2D object) to draw conducting strip and ground plane while FR4 as material to draw rectangular box (3D object), the geometry in Figure 3.4 (a) are thus obtained.

②. Apply boundaries

For solving the electromagnetic field problem of a structure, the boundaries of the materials are needed to assign to 2D (sheet) objects or the surface of the 3D objects. In the case of designing the antenna, the antenna radiation pattern needs to be involved. Therefore, the radiation box is needed and the Radiation Boundary Condition (ABC) or Perfectly Matched Layer (PML) is necessarily assigned for creating the virtual free space. In addition, this virtual free space (i.e. assigned radiation box) has ability to absorb the incident wave at the surface of the box, which cause the standing wave within the box region and therefore decrease the performance of the antenna. PML have better ability to absorb the incident wave than ABC while ABC requires executed time than PML. Particularly, by drawing the radiation box and assigning such box as PML, HFSS will automatically create the PML radiation box. As a result, the radiation box size and the type of radiation condition should be chosen carefully. In the design procedure that will be involved in the following section, Perfect Electric Conductor (PEC) is chosen to assign to the conductor and PML boundary is assigned to the radiation box. As suggested in (Sligar, 2007), the distance of $\lambda / 8$ from the strongly radiating structure and

thickness of $\lambda / 3$ at the lowest frequency of interest are chosen for creating the PML box. In design that will present in the following section, the distant from antenna elements as shown in Figure 3.4 (b) and thickness of PML as shown in Figure 3.4 (c), which are specified for designing the proposed antenna, are 60mm ($\lambda / 8$ at 500MHz) and 160mm ($\lambda / 3$ at 500MHz), respectively.

③. Apply excitation

After assigning boundary condition, the excitation should be applied to the port, as shown at the edge of Figure 3.4 (d), for exciting the antenna. Seven types of excitation are available in HFSS. On the other hand, there are only two excitations that are most commonly used, which are wave port and lumped port. Since wave port cannot apply within the radiation box, lumped port is preferred in this design.

④. Setup solution

Once excitation is assigned to the port, the solution setup need to be performed. In this step, user specify the solution frequency, desired frequency band, the maximum iterative number of adaptive solutions, the convergence criteria, and frequency sweep methodology for obtaining the accuracy result. As using sweep frequency, the center frequency should be chosen to obtain an accuracy result in short period. The solution setup that will be performed in the next section is 1.75GHz solution frequency (center frequency) with 401 points of fast frequency sweep from 0.5GHz to 3GHz, 20 maximum adaptive solution processes with maximum delta of 0.02, and 0.0001 relative residue.

⑤. Solve

After completing the above four steps, the model is now ready to be analyzed and solved. According to the complexity of the model, the time executed in HFSS can take from few second to overnight for running a simulation.

⑥. Post process

When completed the simulation, the field behavior are executed. In HFSS, user can plot the S-parameter as shown in Figure 3.4 (e), VSWR, current distribution, near and far field, and create animation etc.

3.2. Antenna Design

As the rapid development of the wireless communications, a high-performance, low-cost, lightweight, and compact-size antennas are needed to design for integrating with the systems. Monopole antenna is preferred in this design since it contains broad impedance bandwidth, omnidirectional radiation pattern, and high efficiency. Furthermore, due to low-

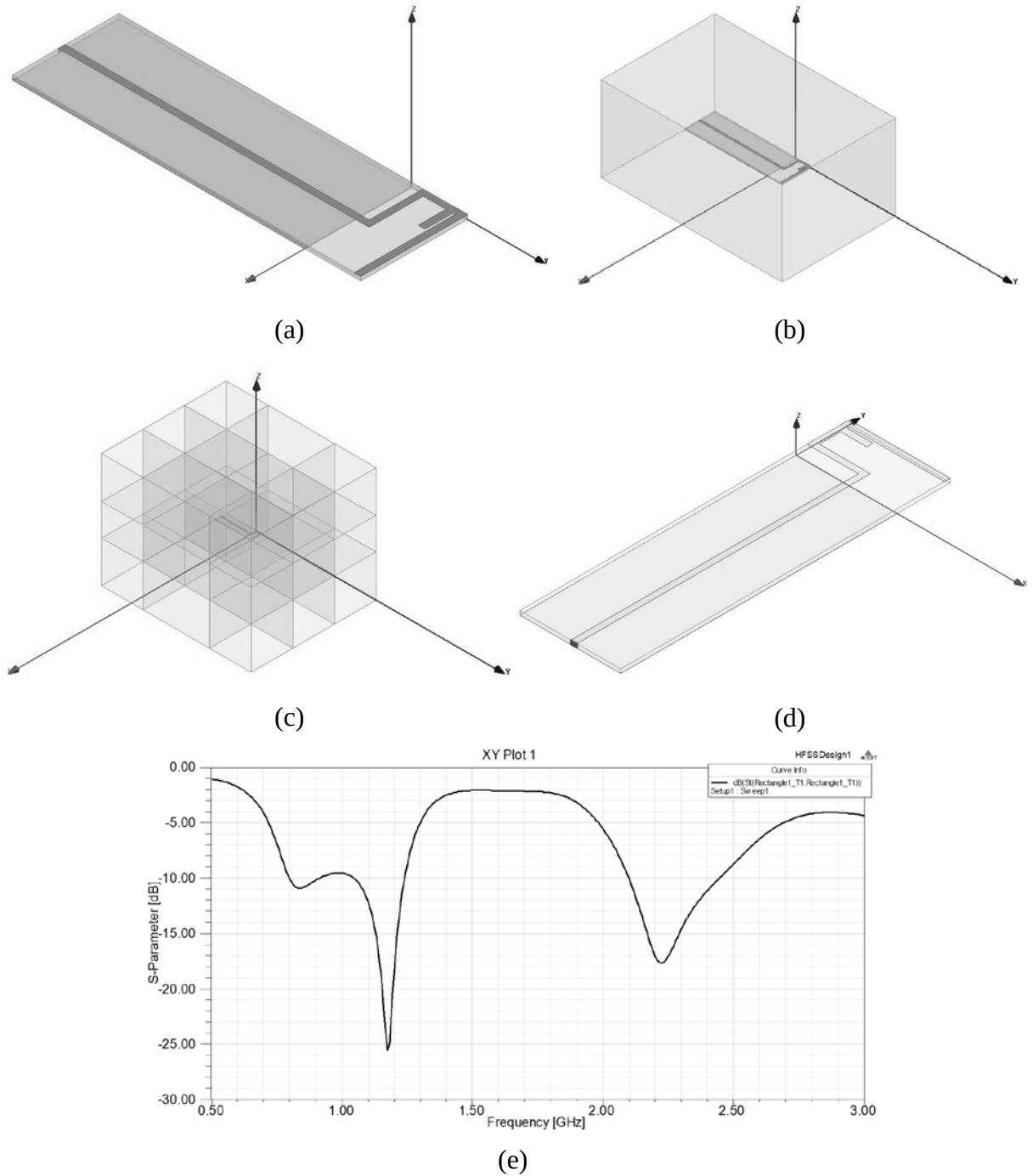


Figure 3.4. HFSS user defined procedures (a) Create geometry (b) Draw radiation box (c) Generate PML for radiation box (d) Assign excitation to port (e) Post process of S-Parameter

cost fabrication, versatile shape, and compact size, planar structure antenna using Printed circuit Board (PCB) technique satisfy the benefit the of the modern mobile terminal. In addition, the available space of mobile terminal for common antenna is typically $30\text{mm} \times 50\text{mm} \times 5\text{mm}$. As a result, miniaturized technique is desired for shrinking the antenna size. There are many techniques that can be utilized for minimizing the size of the printed

monopole antenna like dual band fractal monopole, fractal Inverted-L Antenna (ILA), Planar Inverted-F Antenna (PIFA) (Luo, Pereira, & Salgado, 2014), Inverted-F shape, S-shape and meandered strip antenna (Zhang, Li, Jin, & Wei, 2011). In other words, bent monopole antenna can be useful for optimizing antenna size without increasing of the overall size the antenna, but only a portion of geometries that can be available in designing compact antenna. In following section, antenna which occupy only $22\text{mm} \times 42\text{mm} \times 1.6\text{mm}$ are designed to support the cellular communication which are GSM850 (824 – 894MHz) , GSM900 (890 – 960MHz) , GSM1800 (1710 – 1880MHz) , UMTS (1920 – 2170MHz) , LTE2300 (2305 – 2400MHz) , and LTE2500 (2500 – 2690MHz) bands. Furthermore, this kind of antenna can also hold the frequency band for Zigbee, WLAN, and RFID (2400 – 2484MHz) application. Therefore, it is chosen for designing multiband and wideband printed monopole antenna that will be described in this section. The following section will describe in detail about the 3D model of the antenna that will be generated. In addition, some parametric optimizations are also involved to lead the design to the result validation. In following section, all the frequency bandwidth of antenna is designed to obtain the desired frequency that base on -7.5dB return loss or equivalent to 2.5:1 VSWR.

3.2.1. Antenna and Feed Line Parametric Calculation

Before starting to optimize the antenna parameters using numerous simulator (i.e. HFSS), a quit calculation of the antenna and feed line parameters are necessary.

①. Monopole Antenna's length calculation

The length of antenna can be determined from (Eq. 23). Since the common characteristic impedance of transmission line is 50Ω , antenna input impedance of 50Ω is chosen for matching between the transmission line and transmission line. Since the maximum input impedance of antenna is smaller than 100.265Ω , the third equation of (Eq. 23) is satisfied, therefore

$$R_{in} = 50\Omega = 5.57G^{4.17} \Rightarrow G = e^{\frac{1}{4.17} \ln\left(\frac{50}{5.57}\right)} \simeq 1.69264 \quad (\text{dimensionless})$$

From (Eq. 22), the length of antenna can be computed as

$$G = 1.69264 = kl \Rightarrow l = \frac{1.69264}{2\pi} \lambda = 0.26939\lambda \quad (\text{Eq. 24})$$

which satisfy the condition that l lie between $\lambda / 4$ and 0.3183λ .

In the following design, the above formula will be used for rapid calculation the length of the planar monopole antenna. In addition, it will be seen that a little difference between a quit calculated results and optimized results.

②. Microstrip line's width calculation

As mentioned in previous section, microstrip line is used to excite the antenna. As a result, the dielectric constant, conducting-strip width, and characteristic impedance of microstrip line should be specified. There is the implicit of the formulas for computing these parameters as introduced in (Eq. 18), (Eq. 19), (Eq. 20), and (Eq. 21). In other words, the independence value cannot be determined without supposing a fix value. Therefore, once should specify a fix value and then calculate the desired parameters. If the result is satisfied, then the value is thus be obtained.

Starting from (Eq. 21), Since the thickness of the conducting strip is very thin compare to the high of substrate $t \ll h$, the effective width (or electrical width) $w_{eff}(0)$ of conducting strip is therefore equivalent to the thickness of physical dimension of w .

Considering the physical strip width (or equivalently effective strip width) of the microstrip line as 3mm, the ratio of w_{eff} / h is thus equal to 1.875mm, which is greater than one, in the case that the thickness of substrate is equal to 1.6mm. The effective dielectric constant of microstrip line is therefore computed from (Eq. 19) as

$$\varepsilon_{r,eff}(0) = \varepsilon_{r,eff}(f = 0) = \frac{4.4 + 1}{2} + \frac{4.4 - 1}{2} \left[1 + 12 \frac{1.6}{3} \right]^{-1/2} = 3.32493$$

There characteristic impedance of microstrip line is therefore can be calculated from (Eq. 19) as

$$Z_c(f = 0) = \frac{60}{\sqrt{3.32493}} \ln \left[\frac{8 \times 1.6}{3} + \frac{3}{4 \times 1.6} \right] = 51.16935\Omega$$

which is closely to 50Ω as desired. The same for $w_{eff} = 2.5\text{mm}$, the effective dielectric constant of $\varepsilon_{r,eff}$ and characteristic impedance of Z_c is obtained as 3.32702 and 56.14035Ω , respectively.

To examine whether the microstrip line could support the desired frequency band for exciting the antenna or not, the resonant frequency of the microstrip line should be involved. As presented in previous section, the mentioned microstrip line support only TM field configuration. From (Eq. 17), the resonance frequency of microstrip line in dominant mode

proposed antenna has been miniaturized. In addition, the electromagnetic coupling between these two strips increased the impedance matching over the lower frequency of higher-order resonance frequency. These two strips monopole antenna are fed by microstrip line. Microstrip line feeding method is used in this design since it is facilitated to change its width in order to obtain the desired characteristic impedance for matching the common transmission line to the antenna as the calculation in previous section. The proposed antenna is simulated using HFSS v13.0.2. The simulation that will be processed in HFSS software package below is based on the procedures that have been introduced in section 3.1.2.

From previous section, to obtain approximately 50Ω characteristic impedance of feed line, microstrip line with conducting strip width of 3mm complete this significant. On the other hand, it is also available (in term of impedance matching) to choose the strip width of 2.5mm whose characteristic impedance is 56.14035Ω (as computed from previous section). Moreover, to obtain small dimension of antenna and to fit the antenna width to the microstrip line's strip, 2.5mm width of monopole antenna is therefore chosen. In the following section, the wide of all strips that are taken into account for the design are 2.5mm .

As shown from previous section, optimizing antenna dimensions will increase Q of antenna. This led to narrower the impedance bandwidth as written in the relation of (Eq. 5). It is thus insufficient for designing the wideband antenna. As a result, the antenna dimension should be carefully miniaturized. The proposed antenna whose dimension is $22\text{mm} \times 42\text{mm} \times 1.6\text{mm}$ is realized in this design.

The first step of design is to optimize the length and geometry of F-shape strip to obtain the frequency band that resonance around 900MHz . From (Eq. 24), the length of F-shape strip without additional strip branch (i.e. horizontal- Γ strip) that resonate at 900MHz is

$$l_1 = 0.26939 \times \left(2.99792458 \times 10^8 / 9 \times 10^8 \right) \simeq 89.73\text{mm}$$

Optimizing this parameter in numerous simulation, the total length of 87.75mm is thus obtained. In addition, the higher-order resonance frequency around 2200MHz is also associated with this single horizontal- Γ strip. Moreover, adding an additional branch to the Γ -shape strip increase the impedance matching over the whole frequency band. The position and length of this additional branch is obtained by optimization in numerous simulation.

A single F-shape strip monopole antenna cannot support the frequency band for GSM1800 , GSM1900 and UMTS , which operate at 1710 – 2170MHz band. Therefore, another strip should be added. Another strip that has a form of horizontal- Γ that has a total length around

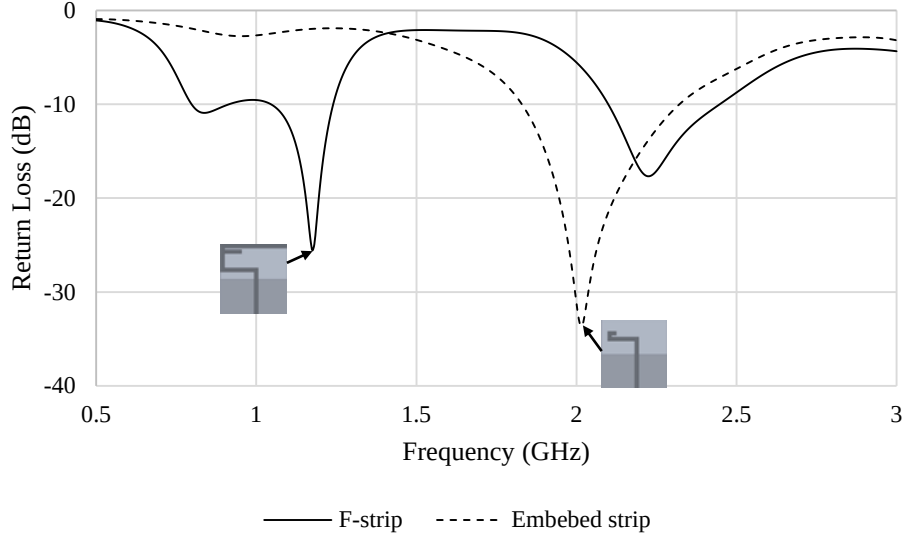


Figure 3.6. Plot of return loss versus frequency of F-shape strip (solid line) and horizontal Γ -shape or embedded strip (dash line) monopole antenna

$$l_2 = 0.26939 \times (2.99792458 \times 10^8 / 20 \times 10^8) \simeq 40\text{mm}$$

The operating frequency of 2000MHz is thus obtained. In addition, for miniaturizing antenna dimension, the horizontal- Γ strip is embedded into the F-shape strip. The position and geometry of horizontal- Γ strip is optimized by simulation to obtain desired frequency band without affecting the operating frequency band of F-shape strip monopole antenna. As a result, the return loss in logarithmic scale versus frequency is obtained in Figure 3.6 by using geometric parameters, which is specified in Table 3.1.

Without additional strip line (ladd), the bandwidth of two-strip monopole antenna is insufficient for LTE2600 frequency band that operate from 2500 to 2690 MHz. Therefore, the additional strip (ladd) is added to the two-strip monopole antenna to increase the frequency band at the higher-order resonance frequency band. The additional strip line of length of 7mm is optimized by numerous simulation. In addition, to obtain sufficient impedance matching bandwidth over the frequency band that listed in Table 1.1, the geometric parameters of proposed antenna in Table 3.1 should be optimized. For the optimization process in the rest of section, without any specification the value of antenna parameters that list in Table 3.1 are assume.

3.2.3. Parametric Optimization

From simulation, increasing the impedance matching bandwidth over the first resonance frequency will decrease the impedance matching over the higher-order resonance

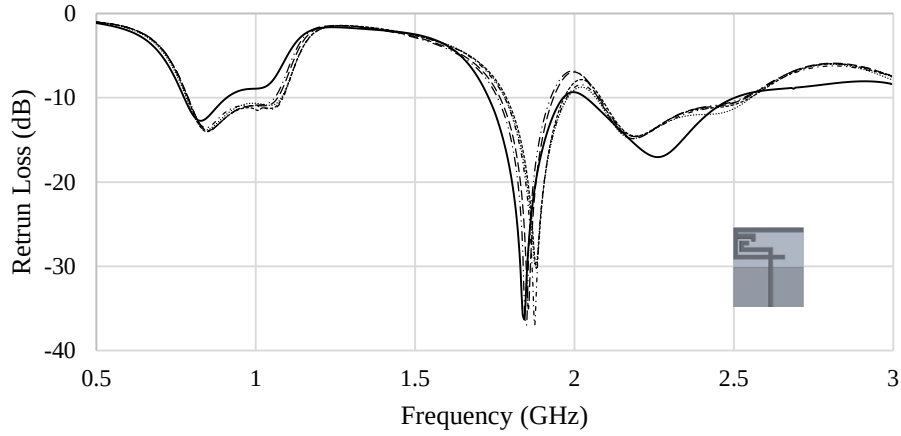


Figure 3.7. Addition and optimization of extended length to obtain the desired impedance matching bandwidth

frequency. Moreover, increasing the length feed gap will increase the high of the antenna. In this case, it is inefficiency to miniaturize the dimension of antenna. Therefore, to obtain the optimal dual-band resonance frequency (i.e. at 900MHz and 2200MHz) with the compact size as the reference operating frequency band and geometry for the whole design, the length of feed gap (lfg) is need to be optimized first. From simulation, the impedance matching bandwidth over the lower of the first resonance frequency increases when the length of feed gap decrease from 1mm to 3.5mm and start to decrease from 3.5 to 4.5mm . On the other hand, the impedance matching bandwidth at the high resonance frequency starts to increase when the length of feed gap increased. As the compact size of proposed antenna is desired, the length of feed gap of 4.5mm is therefore chosen.

It should be noticed that the electromagnetic coupled between two-strip configurations is the very effective method for increasing the bandwidth of compact antenna (Zhang, Li, Jin, & Wei, 2011) and (Li, Ren, Zhao, & Jiao, 2010). In other words, electromagnetic coupling between F-shape strip and horizontal Γ -shape strip of proposed is strongly affect the

Table 3.1. Optimized parameters of proposed antenna

Parameter	Value (mm)	Parameter	Value (mm)
lg	115	ls2	15
wg	42	dst	1.4
h	22	ladd	7
t	1.6	lect	1.5
wf	2.5	lemb	19.85
lfg	4.5	lemb2	1.2
lfg2	1.4	lemb3	5.5
ws	2.5	hb	9
ls	21.25	lb	8.3

performance of the antenna in the entire frequency band. It is thus carefully to optimize this parameter to obtain the desired impedance matching bandwidth. By numerous simulation, the distant between two strips (dst) is thus optimized and recorded as 1.4mm. In addition to optimization of the distant of electromagnetic coupling between two-strip configurations, the extended line is one of the most powerful parameters that could increase the impedance matching bandwidth at the higher-order resonance without affecting the first-order resonance frequency band. The extended length (l_{ext}) from 0.5 to 2.5mm with step size of 0.5mm is simulated by HFSS as shown in Figure 3.7. As a result, the extended length of 1.5mm is the optimal value that can be obtained for the simulation. Moreover, the extended length could lead the design to satisfy frequency requirement band for modern mobile devices. From the operating frequency that listed in Table 3.2, the fractional VSWR bandwidth of antenna at first-order frequency band, which resonate at 900MHz , can be computed from (Eq. 2) as

$$FBW_{v,1} = \frac{\omega_+ - \omega_-}{\omega_0} = \frac{f_{H1} - f_{L1}}{f_{0,1}} = \frac{1061.9 - 744.8}{900} = 35.23\%$$

For the value of fractional VSWR matched bandwidth, antenna Q of 3.9 is obtained from (Eq. 5). By analogous, the high frequency band, which resonate at 2200MHz , can be computed from (Eq. 2) as

$$FBW_{v,2} = \frac{\omega_+ - \omega_-}{\omega_0} = \frac{f_{H3} - f_{L2}}{f_{0,2}} = \frac{3000 - 1695.7}{2200} \simeq 59.28\%$$

Which is correspond to the value of Q of 1.6.

From relationship between the value of Q -factor and fractional bandwidth, it can conclude that the value of Q must be small in order to obtain wide frequency band.

In addition to the mentioned optimized parameters, since the proposed antenna is obtained by two-strip monopole antenna and its ground plane is finite, the size of antenna ground plane is strongly affect the performance of antenna. As confirmed by simulation, the width of ground plane has a little effect to the antenna performance. On the other hand, the

Table 3.2. Operating Frequency band for each value of extended length (VSWR $\leq$ 2.5)

l _{ext} (mm)	f _{L1} (GHz)	f _{H1} (GHz)	f _{L2} (GHz)	f _{H2} (GHz)	f _{L3} (GHz)	f _{H3} (GHz)
0	0.7542	1.1091	1.7416	—	—	2.6503
0.5	0.7578	1.1102	1.7452	—	—	2.6602
1	0.7582	1.1108	1.7372	—	—	2.6620
1.5	0.7448	1.0619	1.6957	—	—	—
2	0.7595	1.0996	1.7265	1.9711	2.0269	2.6512
2.5	0.7598	1.0921	1.7175	1.9617	2.0286	2.6488

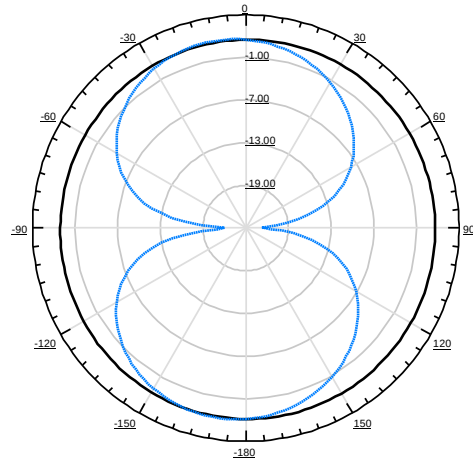
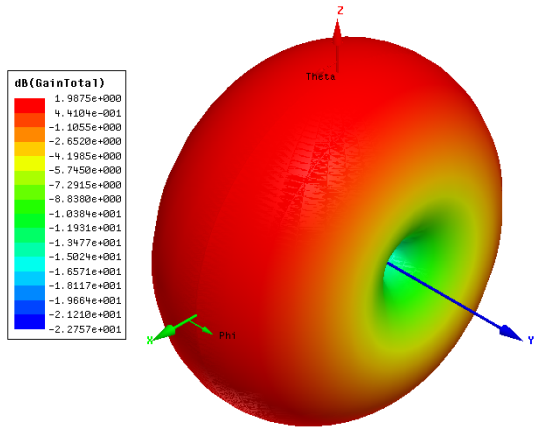
decreasing of the length (l_g) of ground plane deteriorates the impedance matching bandwidth over the low operating frequency band. l_g of 115mm is the optimal value, which is chosen.

3.2.4. Antenna Far Field Solution

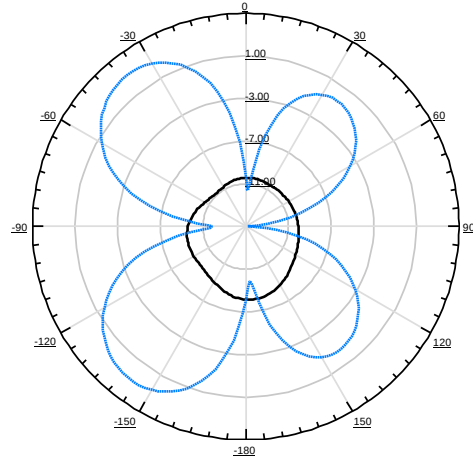
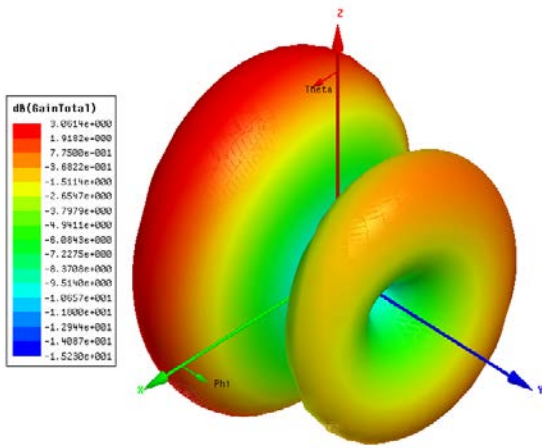
As mentioned from previous section, S-Parameters are taken into account in design for defining desired impedance matching bandwidth. On the other hand, return loss of antenna describe only how much the power is transmitted to antenna over the interested frequency band, but it does not show how well of antenna can radiate the electromagnetic wave into unbounded medium. Therefore, the radiation pattern (i.e. power pattern, field pattern, or gain pattern) and radiation efficiency are the indispensable parameters of the antenna in defining the performance of antenna.

In general, the radiation patterns are expressed in free space coordinate. On the other hand, it is commonly to use a series of 2-dimensional plot of radiation pattern for characterizing the field pattern of antenna in free space, but, for most application, the field pattern in azimuth (x - y plane) and elevation plane (x - z plane) is sufficient for showing the desired information of field pattern. Figure 3.8 shows simulated result of radiation pattern in 3-dimensional (Left) and 2-dimentional (Right) polar plot. The right patterns in blue and dash line show the radiation pattern in elevation or z - x plane while the right patterns in black and solid line show the radiation pattern in azimuth or x - y plane (for more detail about coordinate of antenna, look at the structure of antenna in Figure 3.5). These patterns are expressed in dBi for the total gain patterns, which operate at frequency of 900 MHz , 1850 MHz and 2400 MHz , shown in Figure 3.8 (a), Figure 3.8 (b), and Figure 3.8 (c), respectively. From such pattern, it should be note that the pattern in elevation plane at 900 MHz is seen like the structure of donate (i.e. omnidirectional pattern in this plane). In other words, although the monopole antenna is miniaturized, it also exhibits an omnidirectional (or non-directional) pattern, which is approximate to the radiation pattern of simple conventional monopole antenna.

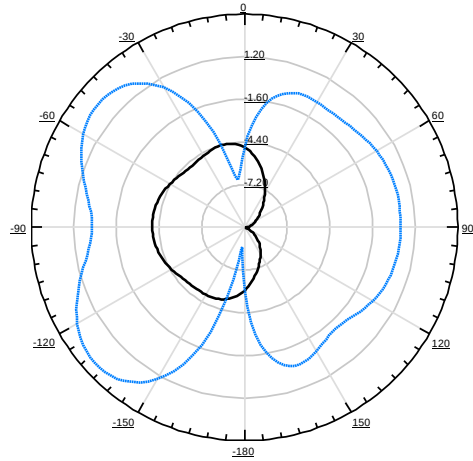
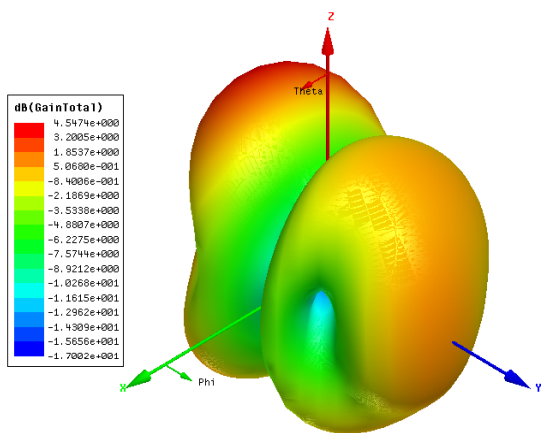
As introduced in section 2.2.2, antenna gain and radiation efficiency are the most powerful parameter for characterizing the performance of antenna. Figure 3.9 show the simulated peak gain and radiation efficiency result at the lower-resonance frequency band of multiband and wideband printed monopole antenna. It can be seen that peak gain from 1.8812dBi to 2.0257dBi and radiation efficiency from 86.73% to 90.67% are obtained at this lower-order resonance frequency band. The same for Figure 3.9 (b) shows the peak gain from 1.4941dBi to 4.5661dBi and radiation efficiency from 70.13% to 84.85% of higher-order resonance frequency of multiband and wideband monopole antenna.



(a)

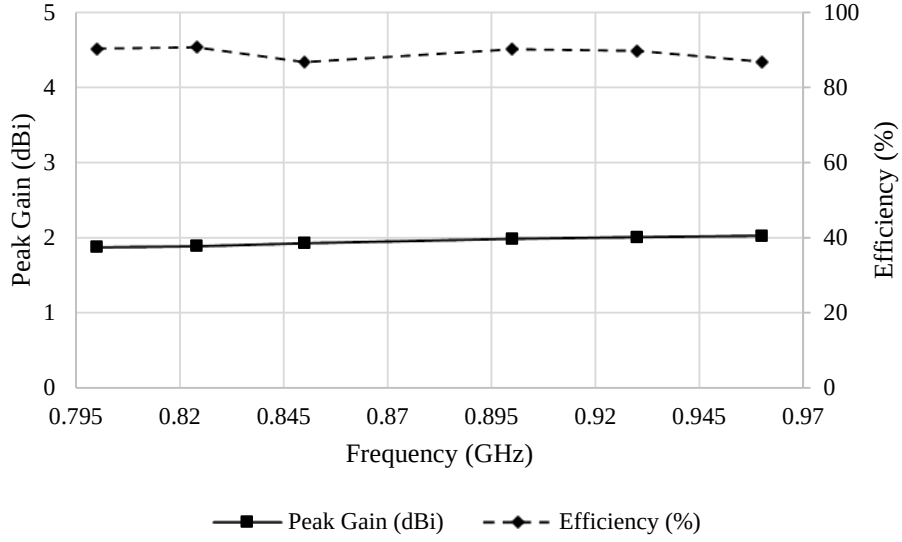


(b)

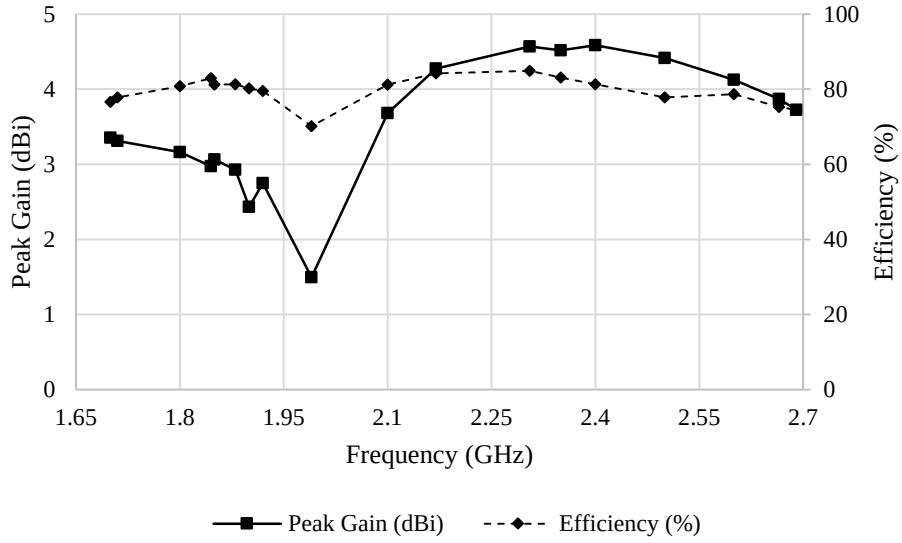


(c)

Figure 3.8. 3-Dimensional total gain pattern (Left) and 2-Dimensional total gain pattern (Right) in dBi at (a) 900 MHz (b) 1850 MHz (c) 2400 MHz



(a)



(b)

Figure 3.9. Simulated Peak Gain (dBi) and radiation efficiency (percentage) values of multiband and wideband printed monopole antenna at (a) lower-order resonance frequency band and (b) higher-order frequency band.

3.2.5. Realized Antenna and Measured Antenna System

To ensure the real performance and characteristic of multiband and wideband printed monopole antenna, a prototype were fabricated and measured. The proposed antenna was fabricated on a FR4 substrate with a thickness of 1.6mm and dielectric constant of 4.4. In addition, both sizes of substrate are etched with conducting sheet. Upper image of Figure 3.10

show the front view of the fabricated antenna, while lower image of Figure 3.10 show back view of the fabricated antenna.

From many antenna measurement literatures, many parameters characterize the performance of antenna. Moreover, As mentioned from previous section, there are many factor, which compose of return loss, VSWR, group velocity and radiation pattern, are used for determination the performance of antenna. For measuring the power, field, and gain pattern of antenna, there are many sophisticated setup of the measurement system. In other words, many equipment and measured environment are desired for realizing the mentioned parameter. For the measured environment, anechoic chamber or large coverage of free space needs to be involved. On the other hand, antenna positioner, power meter, signal generator, VNA, data-acquisition software package, and other connectors, are needed to associate with measurement system. Due to this case, it makes the antenna radiation pattern measurement unrealizable in this project. Therefore, only return loss and voltage-standing wave ratio will be appeared in the following measurement.

On the other hand, to measure the return loss and voltage standing-wave ratio of antenna, vector network analyzer is necessary to be taken into account. Moreover, Agilent E5071C Series Network Analyzer (VNA) is involved in this measurement. By connecting 50Ω SMA connector to the antenna microstrip fed line, the antenna can be measure by connecting one port of VNA to the SMA connector via coaxial cable. For obtaining accuracy result, it should be noted that the connector and coaxial cable must support the frequency, which agrees with the operating frequency range of the antenna.

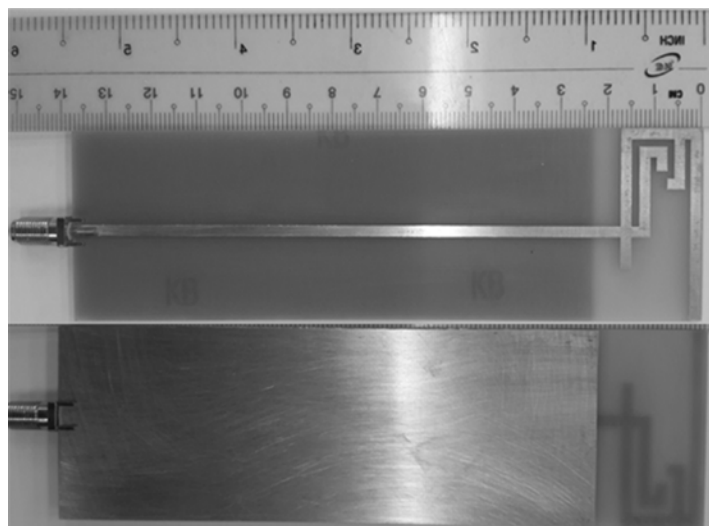
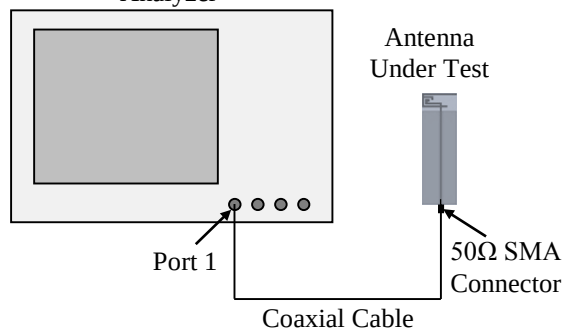


Figure 3.10. Photograph of Front view (upper image) and back view (lower image) of multiband and wideband printed monopole antenna prototype

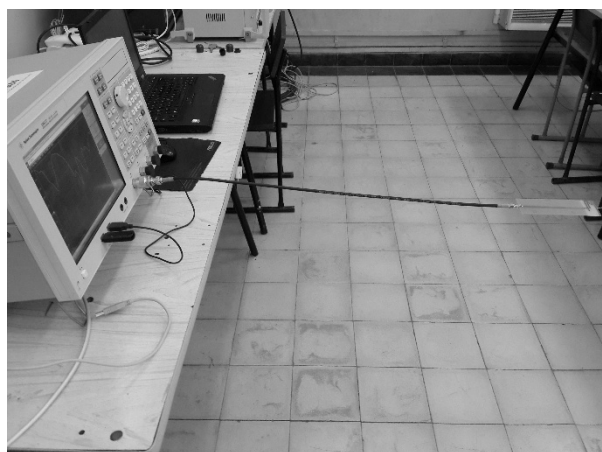
3.3. Vector Network Analyzer Measurement

To make an accuracy measurement, two necessary steps are needed. One of these two steps is calibration. Calibration can reduce some errors that occur during the measurement. Three main errors (directivity, source match, and frequency tracking errors) will be occurred during the measurement and can be reduced by using Electronic calibration kit. It should be noted that the variation of the VNA parameter and measurement environment will can lead to invalidate of the solution. Therefore, for making an accuracy measurement, the VNA parameters and measurement environment should not be varied. The frequency band that is specified before making calibration in this measurement is 2.5GHz (from 0.5GHz to 3GHz). After calibration, the calibration kit is removed from the VNA port and replaced by Antenna Under Test (AUT), which is shown in Figure 3.11. After connecting AUT to VNA and starting to transmit signal, VNA measure the reflection wave and compare with the incident wave. As a result, return loss, VSWR, group delay, antenna impedance can be displayed and recorded. However, to reduce noise error, narrowing IF bandwidth, increasing power, and averaging sweep measurement should be applied. In this measurement, the smoothing of 3.9% (averaging), narrowing IF bandwidth of 70kHz and transmitting power of 0dBm are chosen over the frequency range of 2.5GHz (from 0.5GHz to 3GHz). Finally, we obtain the desired measurement data. Finally, the experimental results of S-Parameter or VSWR can be monitored and recorded.

Agilent E5071C Series Network Analyzer



(a)



(b)

Figure 3.11. Vector Network Analyzer measurement (a) schematic (b) photograph

4. RESULTS AND DISCUSSION

Figure 4.1 and Figure 4.2 show a good agreement between measured (dash line) and simulated (solid line) results of return loss and voltage standing-wave ratio (VSWR). Particularly, it can be observed that two wide operating frequency bands are obtained. Even though there is a little deviation between simulation and measurement, this result also valid due to some avoidable error that occur during measurement and setting simulation parameters. The major errors that occur are caused by experiment processing. Particularly, experimental environment is the most powerful factor that causes the error during the measurement. Since antenna can act as linear and reciprocal device, it can be act as both transmitter and receiver. Without anechoic chamber or free space, the transmitted signal from antenna will reflect back due to the limitation of measurement environment. In this case, the antenna will received the reflected wave and therefore cause the standing wave at the proximity of the antenna. As a result, the antenna measurement performance is thus decrease.

Without anechoic chamber or free space, the antenna measurement is also acceptable in this measurement. At low frequency band which defined by -7.5dB return loss, simulated result of 317.1 MHz ($774.8 - 1061.9\text{ MHz}$) impedance bandwidth (or equivalent to fractional VSWR bandwidth of 35.23% at 900 MHz) is compare to measured result of 225 MHz ($818.8 - 1043\text{ MHz}$) impedance bandwidth (or equivalent to fractional VSWR bandwidth of 25% at 900 MHz). Even though the measured bandwidth result of proposed antenna at low frequency band has narrower bandwidth than simulated result, it still capable to cover the desired frequency ranges. These desired operating frequency are GSM850 ($824 - 890\text{ MHz}$) and GSM900 ($890 - 960\text{ MHz}$). In addition, for RFID/Zigbee based application that operates from $868 - 870\text{ MHz}$ and $902 - 928\text{ MHz}$ are also valid for this operating frequency band.

At high operating frequency, a wide impedance matching bandwidth of both measured and simulated result are observed. In other words, impedance bandwidth of 1304.3 MHz ($1695.7 - 3000\text{ MHz}$) or equivalent to 59.28% at 2200 MHz is confirm by simulation while impedance bandwidth of 1160 MHz ($1635 - 2795\text{ MHz}$) or equivalent to 52.72% is obtained from experiment. In addition, at this operating frequency band, the impedance bandwidth of measured result exhibit a little shifted down compared to simulated result. However, this higher operating frequency band can cover the frequency band for GSM1800 ($1710 - 1880\text{ MHz}$), UMTS ($1920 - 2170\text{ MHz}$), LTE2300 ($2305 - 2400\text{ MHz}$), LTE2600 ($2500 - 2690\text{ MHz}$), WLAN/Wi-Fi ($2400 - 2484\text{ MHz}$), and RFID/Zigbee ($2400 - 2484\text{ MHz}$) applications.

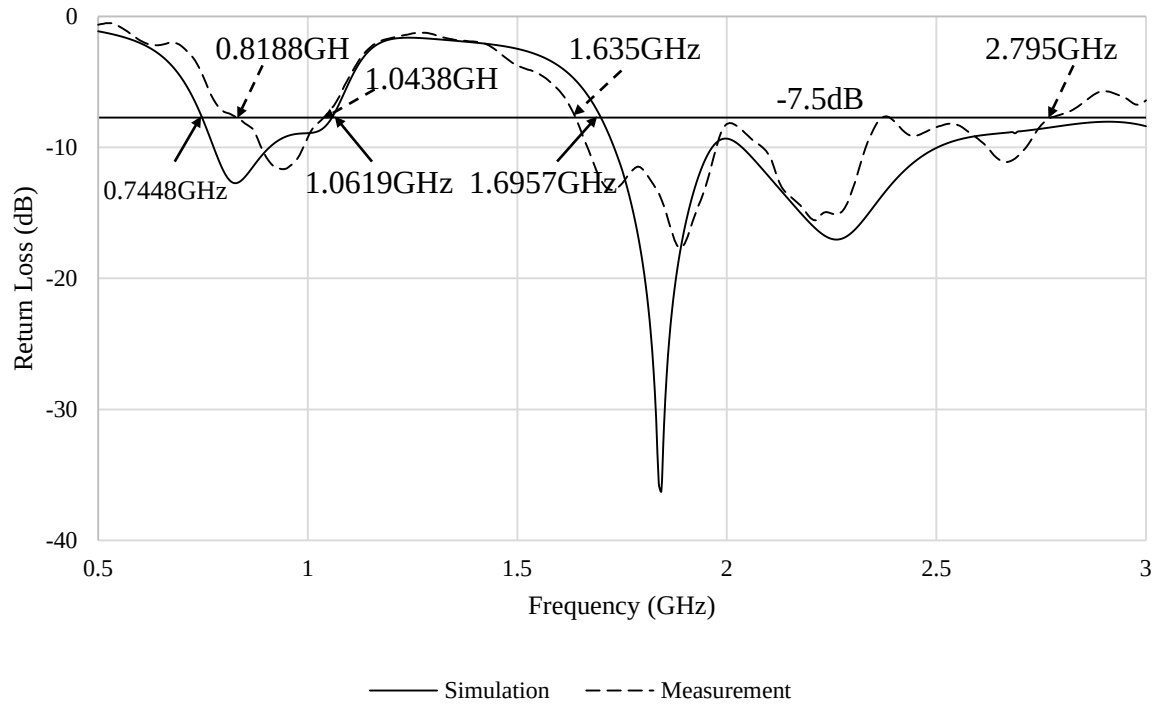


Figure 4.1. Comparison between simulated and measured for return loss of multiband and wideband printed monopole antenna.

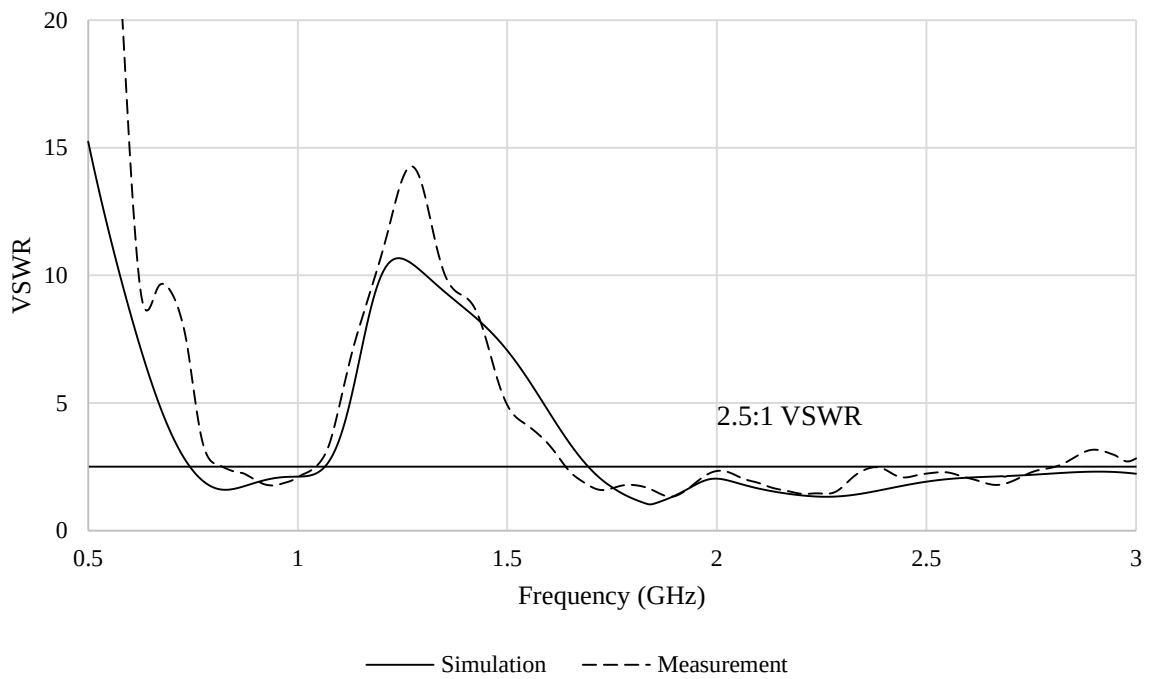


Figure 4.2. Comparison between the simulated and measured result for VSWR of the multiband and wideband printed monopole antenna.

5. CONCLUSIONS

In this study, a novel multiband and wideband printed monopole antenna is designed, built, and measured. Design procedure composes of a quit calculation of antenna and feed line parameters. In addition, to obtain the desired impedance matching bandwidth, High Frequency Structure Simulator, a software package using Finite Element Method, optimize these parameters. Using two strips monopole antenna and keeping those strips in appropriate position to couple mutually each other, the broadband of antenna, which occupies a volume of only $42 \times 22 \text{ mm} \times 1.6 \text{ mm}$, are obtained. The proposed antenna composes of two separated wide frequency bands. The low impedance matching bandwidth covers frequency range of 774.8 – 1061.9 MHz (simulated result) which can support GSM850 and GSM900. Furthermore, the higher frequency band of proposed antenna has a frequency range of 1695.7 – 3000 MHz can cover GSM1800, GSM1900, UMTS, LTE2300, and LTE2600 frequency band. In addition, Zigbee, RFID, Wi-Fi, and WLAN can be operated with this proposed antenna. Antenna gain of 1.8812 – 2.0257 dBi and radiation efficiency of 86.73 – 90.67% are obtained at the lower frequency band while antenna gain of 1.4941 – 4.5661 dBi and radiation efficiency of 70.13 – 84.85% are obtained at the higher frequency band. Thus, the proposed antenna is suitable for mobile terminal as the internal multiband/wideband antenna for wireless communication.

To verify the performance of multiband and wideband printed monopole antenna, a prototype of such antenna was fabricated on a FR4 substrate. There is a good agreement between simulated and experimented results of antenna S-Parameter and VSWR. Simulated result is obtained from High Frequency Structure Simulator while experimented result is obtained from Vector Network Analyzer measurement. Because of the sophisticated and expensive of measurement system, the antenna gain and efficiency cannot be obtained. For more detail about antenna standard measurement and requirement systems, look at *IEEE Test Procedure for Antenna* (IEEE Std 149-1979, 1979).

Even though proposed antenna is successfully designed, further improvement should be involved by choosing more high performance materials. In other words, FR4 substrate that is chosen in this design has high attenuation in term of dielectric loss tangent of 0.02, which make the degradation of antenna performance. To increase the performance to the proposed antenna, the lower dielectric loss tangent like Rogers RT/Duroid substrate with dielectric tangent loss of only 0.0009 should be chosen. In addition to dielectric tangent loss of substrate, the thickness of substrate also decreases the gain and radiation efficiency of antenna. In other words, even though higher thickness of the substrate increases the antenna impedance matching

bandwidth, but it creates the surface loss on the surface of conducting strip. Due to this case, the thin of substrate should be chosen to obtain higher gain and radiation efficiency.

In addition to the choice of the appropriate material, the other design and miniaturized techniques should be included. By adding another strip to the proposed two-strip multiband and wideband printed monopole antenna, another additional operating frequency maybe create without affecting the original frequency band. This technique can be realized by locating the appropriate position between there three strips. On the other hand, the size of ground plane is strongly affect the impedance matching bandwidth at the low frequency band. Therefore, the other methods should be chosen to reduce the effect of ground plane to the antenna.

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APPENDICES: THEORY OF ELECTROMAGNETISM

①. Maxwell's Equation

In differential form, Maxwell's equation can be expressed as:

$$\begin{aligned}\nabla \times \vec{\mathcal{E}} &= -\frac{\partial \vec{\mathcal{B}}}{\partial t} - \vec{\mathcal{M}}, \\ \nabla \times \vec{\mathcal{H}} &= \frac{\partial \vec{\mathcal{D}}}{\partial t} + \vec{\mathcal{J}}, \\ \nabla \cdot \vec{\mathcal{D}} &= q_{ev}, \\ \nabla \cdot \vec{\mathcal{B}} &= q_{mv}.\end{aligned}$$

Where $\vec{\mathcal{E}}$	Electric field intensity (V/m)
$\vec{\mathcal{H}}$	Magnetic field intensity (A/m)
$\vec{\mathcal{D}}$	Electric flux density (C/m ²)
$\vec{\mathcal{B}}$	Magnetic flux density (wb/m ²)
$\vec{\mathcal{J}}$	Electric current density (A/m ²)
$\vec{\mathcal{M}}$	Magnetic current density (V/m ²)
q_{ev}	Electric charge density (C/m ³)
q_{mv}	Magnetic charge density (wb/m ³)

In integral form, Maxwell's equation can be expressed as:

$$\begin{aligned}\oint_C \vec{\mathcal{E}} \cdot d\vec{l} &= -\iint_S \frac{\partial \vec{\mathcal{B}}}{\partial t} \cdot d\vec{s} - \iint_S \vec{\mathcal{M}} \cdot d\vec{s}, \\ \oint_C \vec{\mathcal{H}} \cdot d\vec{l} &= \iint_S \frac{\partial \vec{\mathcal{D}}}{\partial t} \cdot d\vec{s} + \iint_S \vec{\mathcal{J}} \cdot d\vec{s}, \\ \oiint_S \vec{\mathcal{D}} \cdot d\vec{s} &= Q_e, \\ \oiint_S \vec{\mathcal{B}} \cdot d\vec{s} &= Q_m.\end{aligned}$$

②. Boundary Condition

At the surface where two difference media is presented by the source (electric and magnetic source), the derivative of the field vector have no meaning and therefore cannot lead to the electromagnetic field results. In this case, boundary conditions are then applied. Boundary conditions can be expressed as:

$$\begin{aligned}
-\vec{n} \times (\vec{\mathcal{E}}_2 - \vec{\mathcal{E}}_1) &= \vec{\mathcal{M}}_s, \\
\vec{n} \times (\vec{\mathcal{H}}_2 - \vec{\mathcal{H}}_1) &= \vec{\mathcal{J}}_s, \\
\vec{n} \cdot (\vec{\mathcal{D}}_2 - \vec{\mathcal{D}}_1) &= q_{es}, \\
\vec{n} \cdot (\vec{\mathcal{B}}_2 - \vec{\mathcal{B}}_1) &= q_{ms}.
\end{aligned}$$

Where $\vec{\mathcal{E}}_1$	Electric field intensity in medium 1
$\vec{\mathcal{E}}_2$	Electric field intensity in medium 2
$\vec{\mathcal{H}}_1$	Magnetic field intensity in medium 1
$\vec{\mathcal{H}}_2$	Magnetic field intensity in medium 2
$\vec{\mathcal{M}}_s$	Magnetic surface current density (V/m)
$\vec{\mathcal{J}}_s$	Electric surface current density (A/m)
q_{es}	Electric surface charge density (C/m ²)
q_{ms}	Magnetic surface charge density (wb/m ²)
$\vec{n}$	Unity vector normal to the surface boundary point toward from medium 1 to 2.

③.Wave Equation and its Solution

Using four equations of the above Maxwell's equation, current density and field relation in a conductor ($\vec{\mathcal{J}} = \sigma \vec{\mathcal{E}}$), and constitutive relation ($\vec{\mathcal{D}} = \varepsilon_r \varepsilon_0 \vec{\mathcal{E}}$ and $\vec{\mathcal{H}} = \frac{\vec{\mathcal{B}}}{\mu_r \mu_0}$) instantaneous wave equation in source-free ($\vec{\mathcal{M}} = q_{ev} = q_{mv} = 0$) medium can be expressed as:

$$\begin{aligned}
\nabla^2 \vec{\mathcal{E}} &= \mu_r \mu_0 \left(\sigma \frac{\partial \vec{\mathcal{E}}}{\partial t} + \varepsilon_r \varepsilon_0 \frac{\partial^2 \vec{\mathcal{E}}}{\partial t^2} \right) \\
\nabla^2 \vec{\mathcal{H}} &= \mu_r \mu_0 \left(\sigma \frac{\partial \vec{\mathcal{H}}}{\partial t} + \varepsilon_r \varepsilon_0 \frac{\partial^2 \vec{\mathcal{H}}}{\partial t^2} \right)
\end{aligned}$$

Where ε_0	Permittivity in vacuum
ε_r	Relative permittivity of a medium (or Dielectric constant)
μ_0	Permeability in vacuum
μ_r	Relative permeability of a medium

σ Conductivity of a conductor

∇^2 Laplace operator ($\nabla^2 \vec{A} = \vec{a}_x \nabla^2 A_x + \vec{a}_y \nabla^2 A_y + \vec{a}_z \nabla^2 A_z$ in Cartesian coordinate)

In Cartesian coordinate, the general solution of time-harmonic electric field $\vec{E}$ ($\vec{\mathcal{E}}(x, y, z, t) = \text{Re}[\vec{E}(x, y, z)e^{j\omega t}]$) can be expressed as:

$$\vec{E}(x, y, z) = \vec{a}_x E_x(x, y, z) + \vec{a}_y E_y(x, y, z) + \vec{a}_z E_z(x, y, z)$$

By transforming the first expression of instantaneous wave equations into time-harmonic equation and substituting the above expression of $\vec{E}(x, y, z)$ into the transformed equation, time-harmonic wave equation can be expressed as:

$$\begin{aligned} \nabla^2 \vec{E} - \gamma^2 \vec{E} &= \nabla^2 (\vec{a}_x E_x + \vec{a}_y E_y + \vec{a}_z E_z) - \gamma^2 (\vec{a}_x E_x + \vec{a}_y E_y + \vec{a}_z E_z) = 0 \\ \Rightarrow \begin{cases} \nabla^2 E_x(x, y, z) - \gamma^2 E_x(x, y, z) = 0 \\ \nabla^2 E_y(x, y, z) - \gamma^2 E_y(x, y, z) = 0 \\ \nabla^2 E_z(x, y, z) - \gamma^2 E_z(x, y, z) = 0 \end{cases} \end{aligned}$$

where $\gamma^2 = j\omega\mu_0\mu_r(\sigma + j\omega\varepsilon_0\varepsilon_r)$ is the square of the propagation constant.

Using separation-of-variable method, $E_x(x, y, z)$ can be assumed as:

$$E_x(x, y, z) = f(x)g(y)h(z)$$

Substituting the assumed $E_x(x, y, z)$ expression into its wave equation's component, we obtain,

$$gh \frac{\partial^2 f}{\partial x^2} + fh \frac{\partial^2 g}{\partial y^2} + fg \frac{\partial^2 h}{\partial z^2} + \gamma^2 fgh = 0$$

Dividing by fgh , yield

$$\frac{1}{f} \frac{\partial^2 f}{\partial x^2} + \frac{1}{g} \frac{\partial^2 g}{\partial y^2} + \frac{1}{h} \frac{\partial^2 h}{\partial z^2} + \gamma^2 = 0$$

$$\Rightarrow \begin{cases} \frac{1}{f} \frac{\partial^2 f}{\partial x^2} = -\gamma_x^2 \\ \frac{1}{g} \frac{\partial^2 g}{\partial y^2} = -\gamma_y^2 \\ \frac{1}{h} \frac{\partial^2 h}{\partial z^2} = -\gamma_z^2 \end{cases}$$

where $\gamma^2 = \gamma_x^2 + \gamma_y^2 + \gamma_z^2$ is the constraint (dispersion) equation.

By solving the above differential equation, yield

$$\begin{aligned} f_1(x) &= A_1 e^{-\gamma_x x} + B_1 e^{+\gamma_x x} \\ f_2(x) &= C_2 \cosh(\gamma_x x) + D_2 \sinh(\gamma_x x) \\ g_1(y) &= A_2 e^{-\gamma_y y} + C_2 e^{+\gamma_y y} \\ g_2(y) &= C_2 \cosh(\gamma_y y) + D_2 \sinh(\gamma_y y) \\ h_1(x) &= A_3 e^{-\gamma_z z} + B_3 e^{+\gamma_z z} \\ h_2(z) &= C_3 \cosh(\gamma_z z) + D_3 \sinh(\gamma_z z) \end{aligned}$$

By choosing the appropriate function of $f(x)$, $g(y)$ and $h(x)$ which satisfy boundary conditions, the solution of wave equation can be obtained.